Pressure-preserving refinements for simple zeros of the Riemann zeta function

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Abstract

We prove the unconditional lower bound

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \geq \frac{1125000H_{\mathrm{MT}} - 2220}{1120671} = 0.67311752658839043880958574341060586666\ldots,$$

where

$$H_{\mathrm{MT}} = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}}.$$

The proof reuses our previously certified position-weighted seven-point certificate

$$\delta = \frac{39}{10000}, \qquad p = \frac{1}{10^7}(2714, 3733, 3553, 3553, 3733, 2714), \qquad \beta := \sum_{j=1}^{6} p_j = \frac{1}{500},$$

without requiring a new seven-point branch-and-bound computation.

Two refinements are introduced. First, the pressure carried by each relative gap position is retained through the block summation and the shifted-block average. This replaces the coarse global pressure cost $\beta(m-1)$ by the exact mass $\beta(m-6)$. Second, pinching a limiting Gram matrix into the two alternating families of adjacent pairs gives

$$\mathcal{D}(K) \geq \sum_{j=1}^{m-1} w(g_j).$$

Combined with a scalar kernel inequality at $g_0 = 89/100$ and $t = 33/2$, this yields

$$\mathcal{D}(G_B) + P_B \geq \frac{39}{10000}(m-6) - o(1)$$

for $m = 450$. Shifted-block averaging then gives the stated constant. The only new numerical input is the rigorous signed enclosure

$$\frac{171389}{1000000} < k(89/100),$$

certified by an exact-rational verifier; positivity and monotonicity then give $w(89/100) > 2937/100000$. The same verifier checks the pressure mass, the contradiction margin, and the final constant. The complete implication from the archived seven-point certificate and this signed scalar claim to the final epsilon theorem is kernel-checked in Lean. The written mathematical derivation below does not invoke the recent Alpöge–Furman theorem as a premise; the software provenance of the Lean verification is stated separately in Section 9.

1 Introduction and statement of novelty

Let $N(T, 2T)$ denote the number of nontrivial zeros of $\zeta(s)$ with ordinary ordinate in $(T, 2T]$, counted with multiplicity, and let $N_0^s(T, 2T)$ denote the number of simple zeros on the critical line in the same interval. The quantitative study of N_0^s/N goes back to Montgomery's pair-correlation method and the Montgomery–Taylor optimization. Recent work has made the relevant finite-compression and zero-side inertia mechanism unconditional. Alpöge and Furman [5], communicating a proof discovered autonomously by Claude at Anthropic as recorded both by the authors and by Anthropic [6], establish the finite-compression, rank–trace, and inertia route unconditionally. The Montgomery–Taylor clause of their Theorem A gives the optimized simple-zero constant denoted below by H_{MT}. The companion Anthropic Lean artifact retains the historical label "Theorem D" for the corresponding Montgomery–Taylor constant and formal theorem family [9]. Lamzouri [7] subsequently gave a conceptually different unconditional proof of the same 67.25%-level conclusion. For surrounding pair-correlation work see Baluyot–Goldston–Suriajaya–Turnage-Butterbaugh [12, 13] and Goldston–Suriajaya [14, 15]. For the optimized Montgomery–Taylor test family we write

$$H_{\mathrm{MT}} := \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.6725007036794116457\ldots. \tag{1}$$

Two very recent public computational manuscripts are especially close to the present argument. Jo's reproducible draft [19] obtains

$$0.6730085279277797613\ldots$$

using a stability-enhanced rank–trace inequality, a seven-point local functional, and a uniform pressure on its six gaps. A subsequent repository maintained by Rademacher [20], explicitly based on Jo's public artifact, strengthens the uniform seven-point certificate and obtains

$$0.6730213619501665335\ldots.$$

Jo's repository identifies its research draft as generated by GPT-5.6 Sol. Rademacher's repository records Jo's artifact as its upstream provenance and presents the subsequent refinement as AI-generated. We cite the human-maintained artifacts and treat model use as provenance rather than mathematical authorship.

Several contemporaneous public computational artifacts report numerically larger values. Ojha's research repository [10] currently reports a certified candidate bound

$$0.6733127422722459\ldots,$$

and explicitly labels the theorem claim a record candidate pending expert review and end-to-end formalization. Devine's Zenodo preprint [11] reports the unconditional value

$$0.673399.$$

Neither artifact is an input to the present proof. Their proof architectures and validation status differ from ours, so we make no claim here that the numerical value of Theorem 1.1 is the largest publicly reported bound.

The contribution of the present paper is explicit and layered. We do not claim the finite-compression/rank–trace mechanism, the stability-enhanced rank–trace inequality, or the seven-point architecture as new. The position-weighted seven-point certificate itself is an earlier contribution of the present project, frozen in the reproducibility artifact [21]; that version already introduced the nonuniform pressure vector

$$p = \frac{1}{10^7}(2714, 3733, 3553, 3553, 3733, 2714), \qquad \beta := \sum_{j=1}^{6} p_j = \frac{1}{500}, \tag{2}$$

for which the seven-point functional satisfies

$$\mathcal{F}_p(g_1, \ldots, g_6) \geq \frac{39}{10000} \qquad (g_j \geq 0). \tag{3}$$

The new argument has two stages. First, rather than replacing the accumulated position-dependent pressure inside a block by $\beta\,\mathrm{span}(B)$, we retain its exact coefficients until the final average over shifted blocks. This reduces the global pressure mass from the coarse $\beta(m-1)$ to the exact $\beta(m-6)$. Already at $m = 262$ this gives

$$\frac{655000 H_{\mathrm{MT}} - 1280}{652504} = 0.67311152255007575119235 86\ldots.$$

Second, an adjacent-pair pinching argument supplies an independent lower bound for the Gram defect in terms of the consecutive-gap kernel energies. Together with one rigorously enclosed scalar value of the Montgomery–Taylor kernel, this permits $m = 450$ and yields Theorem 1.1. No new seven-point local certificate is required.

Theorem 1.1 (Main theorem)**.** *With*

$$H_{\mathrm{MT}} = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}},$$

one has unconditionally

$$\liminf_{T \to \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{1125000 H_{\mathrm{MT}} - 2220}{1120671} = 0.6731175265883904388095857434106058666\ldots.$$

A second contribution is expository and logical rather than a separate priority claim: we reconstruct the analytic/Gabor package needed by the proof from classical theorems and explicitly identify the common finite compression on its zero and prime sides. Consequently, Theorem 1.1 does not invoke the Montgomery–Taylor clause of Theorem A of Alpöge–Furman [5], or any other theorem of that paper, as a mathematical premise. This is a statement about the logical derivation presented in the manuscript, not a claim of clean-room software independence: the Lean verification has the distinct provenance recorded in Section 9.

Priority and provenance convention. The priority statements above concern the public artifacts inspected while preparing this version. The proof discovered by Claude and verified and communicated by Alpöge–Furman supplies the immediate unconditional finite-compression/rank–trace comparison point [5, 6]. Jo's repository is the public upstream artifact for the stability-enhanced seven-point draft used here; Rademacher's repository records that provenance and strengthens its uniform finite certificate. The position-weighted seven-point certificate (2)–(3) is frozen in our earlier artifact [21]. The new claims of the present version are restricted to the exact pressure-preserving global accounting, the adjacent-pair defect lemma, the scalar kernel enclosure used with that lemma, and the resulting constant. The seven-point branch-and-bound certificate is reused unchanged.

Component	Provenance used in this paper
Classical analytic inputs	Weil explicit formula, Montgomery–Taylor optimization, Riemann–von Mangoldt, PNT, Poisson summation, Montgomery–Vaughan and standard matrix analysis
Finite compression, zero-side inertia and rank–trace route	Proof discovered by Claude (Anthropic), verified and communicated by Alpöge–Furman [5, 6]; not imported as a premise of Theorem 1.1
Stability-enhanced rank–trace and seven-point architecture	Jo [19]
Uniform seven-point strengthening	Rademacher [20]
Position-weighted seven-point certificate 39/10000	Hurtado, earlier frozen artifact [21]
Exact pressure preservation, adjacent-pair pinching, scalar enclosure and $m = 450$ assembly	Present paper
Lean base library	Anthropic `formal-math/zeta23`, pinned commit recorded in Section 9 [9]
Lean v17 refinement layer	`HurtadoZeta23` modules in the present repository

2 The common finite compression and its two representations

Put

$$\ell = \log \frac{T}{2\pi}, \quad L = \ell, \quad h = \frac{2\pi}{L}, \quad d = \left\lfloor \frac{T}{h} \right\rfloor, \quad \tau_j = T + jh \quad (0 \le j < d).$$

Let $\phi = \phi_L \in C_c^2(\mathbb{R})$ be the real, even Montgomery–Taylor window supported on $[-L/2, L/2]$, and define

$$\widehat{\phi}(z) := \int_{\mathbb{R}} \phi(u) e^{-izu}\, du, \qquad \mathcal{F}_+ f(z) := \int_{\mathbb{R}} f(u) e^{izu}\, du.$$

Set

$$f_j(u) := \phi(u) e^{-i\tau_j u}. \tag{4}$$

Since ϕ is even, $\widehat{\phi}$ is even entire, hence

$$F_j(z) := \mathcal{F}_+ f_j(z) = \widehat{\phi}(\tau_j - z) = \widehat{\phi}(z - \tau_j). \tag{5}$$

For a zero $\rho = \beta + i\gamma$ put

$$z_\rho := -i\left(\rho - \frac{1}{2}\right) = \gamma - i\left(\beta - \frac{1}{2}\right). \tag{6}$$

For the horizontal partner $\rho^\sharp := 1 - \overline{\rho}$,

$$z_{\rho^\sharp} = \overline{z_\rho}. \tag{7}$$

Let W denote the Hermitian Weil form in the normalization of Appendix C. Define the single common finite compression

$$G_{jk} := W(f_j, f_k), \qquad 0 \le j, k < d. \tag{8}$$

For $c = (c_0, \ldots, c_{d-1})^t$, put

$$f_c := \sum_{j=0}^{d-1} \overline{c_j}\, f_j, \qquad \Lambda_\rho(c) := \mathcal{F}_+ f_c(z_\rho) = \sum_{j=0}^{d-1} \overline{c_j}\, F_j(z_\rho).$$

With this convention, which is chosen only to match the standard matrix quadratic form, one has $c^*Gc = W(f_c, f_c)$ exactly.

Proposition 2.1 (Two representations of the common compression)**.** *Let $X = e^L = T/(2\pi)$ and let $\nu_X = \mu + P_X + \Pi_X$ be the density derived from the classical Weil explicit formula in Proposition C.1. Then:*

(a) Spectral representation.

$$G_{jk} = \int_{\mathbb{R}} \widehat{\phi}(\tau - \tau_j)\overline{\widehat{\phi}(\tau - \tau_k)}\nu_X(\tau)\, d\tau. \tag{9}$$

(b) Zero-side representation. *For every $c \in \mathbb{C}^d$,*

$$c^*Gc = \sum_{\rho} m_\rho \, \Lambda_\rho(c)\, \overline{\Lambda_{\rho^\sharp}(c)}. \tag{10}$$

Grouping under $\rho \leftrightarrow \rho^\sharp$ gives

$$\begin{aligned} c^*Gc = &\sum_{\rho=\rho^\sharp} m_\rho |\Lambda_\rho(c)|^2 \\ &+ \sum_{\{\rho,\rho^\sharp\},\, \rho\neq\rho^\sharp} 2m_\rho \Re\Big(\Lambda_\rho(c)\overline{\Lambda_{\rho^\sharp}(c)}\Big)\,. \end{aligned} \tag{11}$$

(c) Interior/exterior decomposition. *Let $I' = (T - T^{1/2}, 2T + T^{1/2}]$. Define A by restricting the zero-side sum to zeros with ordinary ordinates in I', and E by the complementary sum. Since $\rho \mapsto \rho^\sharp$ preserves ordinary ordinate,*

$$G = A + E \tag{12}$$

entrywise.

(d) Normalized evaluation vectors. *Put*

$$a = \frac{1}{L}\int_{\mathbb{R}} \phi(u)^2\, du, \qquad \widehat{A} = \frac{A}{aL^2}.$$

For a simple critical-line zero ρ in I', define

$$v_\rho := \frac{1}{\sqrt{a}\, L}\big(F_0(\gamma), \ldots, F_{d-1}(\gamma)\big)^t.$$

Then $\Lambda_\rho(c)/(\sqrt{a}\, L) = c^* v_\rho$, and

$$\|v_\rho\|^2 = \frac{1}{aL^2}\sum_{j=0}^{d-1} |\widehat{\phi}(\gamma - \tau_j)|^2 \le 1. \tag{13}$$

Thus

$$P_1 := \sum_{\rho\in\mathcal{S}_1} v_\rho v_\rho^*$$

is exactly the normalized simple-critical-line part of $\widehat{A}$, and $Q' := \widehat{A} - P_1$ is the pull-back of the remaining zero-side blocks.

Proof. Proposition C.1 and (5) give (9). The zero-side form of the same Weil identity is

$$W(f,g) = \sum_{\rho} m_\rho \, \mathcal{F}_+ f(z_\rho) \, \overline{\mathcal{F}_+ g(\overline{z_\rho})}.$$

By the coefficient convention above, $c^* G c = W(f_c, f_c)$. Using (7) with $f = g = f_c$ therefore proves (10); grouping paired zeros proves (11). Splitting this same zero sum by whether the ordinary ordinate lies in I' proves (12).

For a critical-line zero, $z_\rho = \gamma \in \mathbb{R}$. Extending the finite sum in (13) to the full translated critical lattice and applying Lemma D.2 at equal arguments (translation of the lattice changes only nonzero dual phases, whose alias terms vanish by exact support) gives

$$\sum_{j \in \mathbb{Z}} |\widehat{\phi}(\gamma - \tau_j)|^2 = aL^2.$$

Hence the finite sum is at most aL^2. Finally, (11) identifies the simple part with $\sum v_\rho v_\rho^*$ and the remainder with the pull-back of the multiple and off-line blocks. □

Remark 2.2 (Dependency graph)**.** *Input I uses the zero-side representation above, Input II uses $G = A + E$, Input III uses the spectral representation, and Input IV supplies the critical-lattice identity used in* (13)*. Thus all four modules refer to one test family and one matrix.*

3 Analytic/Gabor package and the stability refinement

Put

$$\ell := \log \frac{T}{2\pi}, \qquad I = (T, 2T], \qquad D_0 = T^{1/2}, \qquad I' = (T - D_0, 2T + D_0],$$

and set $L = \ell$. Let ϕ be the real, even, compactly tapered Montgomery–Taylor window used below, and define

$$a := \frac{1}{L} \int_{\mathbb{R}} \phi(u)^2 \, du, \qquad \widehat{G} := \frac{G}{aL^2}, \qquad \widehat{A} := \frac{A}{aL^2}.$$

The analytic work needed by the stability argument is summarized in the next proposition. Its four assertions are derived in Appendices A–D from the trust base stated below.

Proposition 3.1 (Analytic/Gabor package)**.** *Let the zeros in I' be partitioned into simple critical-line zeros $\mathcal{S}_1$, distinct multiple critical-line zeros $\mathcal{S}_2$, and unordered horizontal pairs*

$$\{\rho, \rho^\sharp\}, \qquad \rho^\sharp = 1 - \overline{\rho},$$

off the critical line. Write $s_1 = \#\mathcal{S}_1$, $s_2 = \#\mathcal{S}_2$ and $p = \#\mathcal{P}$. Then the following hold.

(I) Zero-side inertia.

$$N(I') \geq s_1 + 2s_2 + 2p. \tag{14}$$

For the normalized sampled vectors v_ρ associated with $\rho \in \mathcal{S}_1$, let V have these vectors as columns and put

$$P_1 = VV^*, \qquad M = V^* V, \qquad Q' = \widehat{A} - P_1.$$

Then

$$n_+(Q') \leq s_2 + p. \tag{15}$$

(II) Exterior-zero tail. *Writing $\widehat{A} = \widehat{G} - \widehat{E}$,*

$$4 \operatorname{tr} \widehat{A} - \|\widehat{A}\|_{\mathrm{F}}^2 \geq 4 \operatorname{tr} \widehat{G} - \|\widehat{G}\|_{\mathrm{F}}^2 - o(N(T, 2T)). \tag{16}$$

In fact the proof gives $\|\widehat{E}\|_1 \ll T^{-1/2}$.

(III) Optimized first and second moments.

$$\operatorname{tr}\widehat{G} = N(T,2T)(1+o(1)), \qquad \|\widehat{G}\|_{\mathrm{F}}^2 = \left(\frac{1}{c_1^*} + o(1)\right) N(T,2T), \tag{17}$$

where

$$\frac{1}{c_1^*} = \frac{1}{2} + \frac{1}{\sqrt{2}}\cot\frac{1}{\sqrt{2}}, \qquad 2 - \frac{1}{c_1^*} = H_{\mathrm{MT}}. \tag{18}$$

(IV) Critical-lattice Poisson–Gabor limit. *At mesh* $h = 2\pi/L$, *Poisson summation gives the infinite-grid identity without nonzero aliases. The tapered Fourier decay controls finite-grid truncation, and the normalized overlap kernel converges to the compact Montgomery–Taylor kernel used in Section 4. The integrated second-moment end effect is*

$$O\Big(L\ell \log L\,(\ell^2 + X)\Big), \qquad X = e^L \asymp T,$$

which is $o(TL\ell^2)$ *at* $L = \ell$.

Proof. Assertion (I) is Theorem A.5; assertion (II) is Theorem B.6; assertion (III) is proved in Appendix C, culminating in (95); and assertion (IV) is Theorem D.1. These appendices use only the classical trust base listed in Section 9. □

The stability-enhanced rank–trace inequality used next is due to Jo [19]; we include the proof to fix the normalization and the precise defect functional used in the present argument.

Define the convex function

$$\Psi(t) = \begin{cases} (t-1)^2, & 0 \le t \le 2, \\ 2t-3, & t \ge 2, \end{cases} \qquad \mathcal{D}(M) := \operatorname{tr}\Psi(M) \tag{19}$$

for $M \succeq 0$.

Lemma 3.2 (Stability-enhanced rank–trace inequality)**.** *Let* V *be a* $d \times r$ *matrix whose columns have norm at most one. Put* $P = VV^* \succeq 0$ *and* $M = V^*V$. *If* Q *is Hermitian and* $n_+(Q) \le b$, *then*

$$\|P+Q\|_{\mathrm{F}}^2 \ge 4\operatorname{tr}(P+Q) - 3r - 4b + \mathcal{D}(M). \tag{20}$$

Proof. Write $Q = Q_+ - Q_-$ for its positive and negative parts. They are positive semidefinite, have orthogonal supports, and $\operatorname{rank} Q_+ \le b$. Since $\operatorname{tr}((P-Q_-)Q_+) = \operatorname{tr}(PQ_+) \ge 0$,

$$\|P+Q\|_{\mathrm{F}}^2 \ge \|P-Q_-\|_{\mathrm{F}}^2 + \|Q_+\|_{\mathrm{F}}^2.$$

If $q > 0$, then $q^2 \ge 4q - 4$; summing over the at most b positive eigenvalues of Q gives

$$\|Q_+\|_{\mathrm{F}}^2 \ge 4\operatorname{tr}Q_+ - 4b. \tag{21}$$

Let $p_1 \ge \cdots \ge p_r \ge 0$ be the eigenvalues of $M = V^*V$ (counted with multiplicity, hence including any zeros). The nonzero eigenvalues of $P = VV^*$ are the same; extend the eigenvalue list of P by zeros as necessary. Likewise write $n_1 \ge n_2 \ge \cdots \ge 0$ for the eigenvalues of Q_- and extend either list by zeros to a common length. Von Neumann's trace inequality [8] then implies

$$\operatorname{tr}(PQ_-) \le \sum_{i \le r} p_i n_i.$$

Consequently

$$\|P - Q_-\|_{\mathrm{F}}^2 + 4\,\mathrm{tr}\,Q_- \geq \sum_{i \leq r} \big((p_i - n_i)^2 + 4n_i\big).$$

For fixed $p \geq 0$,

$$\min_{n \geq 0} \big((p-n)^2 + 4n\big) = \begin{cases} p^2, & 0 \leq p \leq 2, \\ 4p - 4, & p \geq 2, \end{cases} = 2p - 1 + \Psi(p).$$

Therefore

$$\|P - Q_-\|_{\mathrm{F}}^2 \geq 2\,\mathrm{tr}\,P - r + \mathcal{D}(M) - 4\,\mathrm{tr}\,Q_-. \tag{22}$$

Combining (21) and (22),

$$\|P + Q\|_{\mathrm{F}}^2 \geq 4\,\mathrm{tr}(P+Q) - 2\,\mathrm{tr}\,P - r - 4b + \mathcal{D}(M).$$

Finally $\mathrm{tr}\,P$ is the sum of the squared column norms of V, so $\mathrm{tr}\,P \leq r$, proving (20). ☐

Before stating the global consequence, define the retained central set

$$\mathcal{S}^\circ := \{\rho \in \mathcal{S}_1 : T < \gamma_\rho \leq 2T \text{ and } \rho \text{ lies outside the two boundary strips of Section 4}\}. \tag{23}$$

Let $S^\circ := \#\mathcal{S}^\circ$ and let

$$M^\circ := \big(\langle v_\rho, v_{\rho'}\rangle\big)_{\rho,\rho' \in \mathcal{S}^\circ}. \tag{24}$$

Thus M° is a principal submatrix of M and contains only simple critical-line zeros whose ordinates lie in $(T, 2T]$ and for which the compact kernel approximation is uniform.

Corollary 3.3 (Global Gram-defect inequality)**.** *Then*

$$N_0^s(T, 2T) \geq H_{\mathrm{MT}} N(T, 2T) + \mathcal{D}(M^\circ) - o(N(T, 2T)). \tag{25}$$

Proof. Apply Lemma 3.2 to $\widehat{A} = P_1 + Q'$ with $r = s_1$ and $b = s_2 + p$. By (14) and (15),

$$\begin{aligned} \|\widehat{A}\|_{\mathrm{F}}^2 &\geq 4\,\mathrm{tr}\,\widehat{A} - 3s_1 - 4s_2 - 4p + \mathcal{D}(M) \\ &\geq 4\,\mathrm{tr}\,\widehat{A} - s_1 - 2N(I') + \mathcal{D}(M). \end{aligned}$$

Hence

$$s_1 \geq 4\,\mathrm{tr}\,\widehat{A} - \|\widehat{A}\|_{\mathrm{F}}^2 - 2N(I') + \mathcal{D}(M). \tag{26}$$

Apply the independently proved tail estimate (16). Pinching M to the retained central principal block and its complement does not increase $\mathrm{tr}\,\Psi$: by the standard pinching/majorization principle for Hermitian matrices [8], pinching is a random-unitary average and the spectral trace functional $X \mapsto \mathrm{tr}\,\Psi(X)$ is convex and unitarily invariant. No operator convexity of Ψ is required. Thus $\mathcal{D}(M) \geq \mathcal{D}(M^\circ)$. Moreover,

$$N(I') = N(T, 2T) + o(N(T, 2T)).$$

Because s_1 counts simple critical-line zeros in the enlarged interval I', whereas $N_0^s(T, 2T)$ counts only those in I, we also use

$$s_1 \leq N_0^s(T, 2T) + N(I' \setminus I). \tag{27}$$

The standard local zero-count estimate gives

$$N(I' \setminus I) \ll D_0 \log T = T^{1/2} \log T = o(N(T, 2T)),$$

and hence

$$N_0^s(T, 2T) \geq s_1 - o(N(T, 2T)). \tag{28}$$

Substitution of (16), (17), and (18) into (26), followed by (28), gives (25). ☐

4 The limiting overlap kernel

Put

$$h := \frac{2\pi}{L}, \qquad x_\rho := \frac{\gamma_\rho - T}{h} = \frac{L(\gamma_\rho - T)}{2\pi}.$$

Define

$$K(x) := \int_{-1/2}^{1/2} \cos(\sqrt{2}\,t)\cos(2\pi x t)\,dt \tag{29}$$

$$= \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}}, \tag{30}$$

$$k(x) := \frac{K(x)}{K(0)}, \qquad w(x) := k(x)^2, \qquad K(0) = \sqrt{2}\sin(1/\sqrt{2}). \tag{31}$$

The apparent singularities in (30) are removable.

Lemma 4.1 (Compact overlap limit). *Fix $R_0 > 0$. Delete simple zeros lying in boundary strips of normalized width L^2 at the two ends of the sampling grid. The number deleted is $o(N(T, 2T))$. Uniformly for retained simple zeros satisfying $|x_\rho - x_{\rho'}| \le R_0$,*

$$\langle v_\rho, v_{\rho'}\rangle = k(x_\rho - x_{\rho'}) + o(1). \tag{32}$$

Proof. By Appendix D, the independently proved Poisson–Gabor sampling identity, applied to the full grid, gives

$$\frac{1}{aL^2}\sum_{j\in\mathbb{Z}} \widehat{\phi}(\gamma - \tau_j)\widehat{\phi}(\gamma' - \tau_j) = \frac{\Phi(\gamma - \gamma')}{aL}, \qquad \Phi = \widehat{\phi^2}.$$

We record explicitly the pointwise truncation estimate needed here. The Fourier decay of the fixed-width tapered window gives, uniformly in L,

$$|\widehat{\phi}(r)| \ll \min\left\{L, \frac{1}{|r|}, \frac{1}{r^2}\right\}.$$

Suppose that both points lie at normalized distance at least L^2 from the finite grid ends and that $|x_\rho - x_{\rho'}| \le R_0$. If an omitted grid index lies $n \ge L^2$ grid steps beyond the nearer end, then

$$|\gamma - \tau_j| \asymp \frac{n}{L}, \qquad |\gamma' - \tau_j| \asymp \frac{n}{L}$$

uniformly for fixed R_0. Hence

$$|\widehat{\phi}(\gamma - \tau_j)\widehat{\phi}(\gamma' - \tau_j)| \ll \frac{L^4}{n^4}.$$

Summing the two omitted tails gives

$$\sum_{n \ge L^2} \frac{L^4}{n^4} \ll L^{-2}.$$

For the optimized window, dominated convergence also gives

$$a = \frac{1}{L}\int_{\mathbb{R}} \phi(u)^2\,du = K(0) + o(1), \qquad K(0) = \sqrt{2}\sin(1/\sqrt{2}) > 0,$$

so in particular $a \asymp 1$. After multiplication by the Gram normalization $1/(aL^2)$, the omitted tails therefore contribute $O(L^{-4}) = o(1)$, uniformly for bounded normalized separation. Thus the full-grid Poisson–Gabor identity may be replaced by the finite sampled inner product with a uniform $o(1)$ error.

For $\gamma - \gamma' = hx$ and fixed x, substitute $u = Lt$:

$$\frac{\Phi(hx)}{aL} = \frac{\displaystyle\int_{-1/2}^{1/2} \phi(Lt)^2 e^{-2\pi ixt}\, dt}{\displaystyle\int_{-1/2}^{1/2} \phi(Lt)^2\, dt}.$$

The squared tapered window converges in L^1 to $\cos(\sqrt{2}\,t)\mathbf{1}_{[-1/2,1/2]}(t)$; the rescaled taper occupies length $O(L^{-1})$. Hence the last quotient converges to $K(x)/K(0)$, uniformly for x in compact sets. Combining this with the finite-grid tail bound proves (32).

Each deleted boundary strip has ordinate length $O(L)$, and the standard local zero-count estimate $N(t+1) - N(t) \ll \log(t+3)$ shows that only $O(L^2) = o(N)$ zeros are removed. $\square$

Lemma 4.2 (Block defect)**.** *If $G \succeq 0$ is Hermitian, then*

$$\operatorname{tr}\Psi(G) \ge \min\left\{1,\ 2\sum_{i<j} |G_{ij}|^2\right\}. \tag{33}$$

Proof. If all eigenvalues of G lie in $[0, 2]$, then $\Psi(G) = (G - I)^2$ and therefore

$$\operatorname{tr}\Psi(G) = \|G - I\|_{\mathrm{F}}^2 = \sum_i (G_{ii} - 1)^2 + 2\sum_{i<j} |G_{ij}|^2 \ge 2\sum_{i<j} |G_{ij}|^2.$$

If some eigenvalue $\lambda > 2$, then its contribution is $\Psi(\lambda) = 2\lambda - 3 > 1$, while all other contributions are nonnegative. This proves (33). $\square$

5 Position-weighted seven-point stability

Let $g_1, \dots, g_6 \ge 0$ be six consecutive gaps. For $1 \le r \le 6$ and $1 \le i \le 7 - r$, write

$$s_{i,r} = g_i + \cdots + g_{i+r-1}$$

and define

$$\mathcal{E}_7(g) := \sum_{r=1}^{6} \frac{2}{7-r} \sum_{i=1}^{7-r} w(s_{i,r}). \tag{34}$$

For $p_j \ge 0$ put

$$\mathcal{F}_p(g) := \sum_{j=1}^{6} p_j g_j + \mathcal{E}_7(g). \tag{35}$$

Proposition 5.1 (Exact pressure redistribution)**.** *Assume*

$$\mathcal{F}_p(g) \ge \delta \qquad (g \in \mathbb{R}^6_{\ge 0}), \qquad \beta := \sum_{j=1}^{6} p_j.$$

Let $y_1 < \cdots < y_m$, $m \geq 7$, and put

$$g_r := y_{r+1} - y_r \qquad (1 \leq r \leq m-1).$$

Define

$$q_r := \sum_{a=1}^{m-6} \sum_{j=1}^{6} p_j \, \mathbf{1}_{\{r=a+j-1\}}, \qquad P_B := \sum_{r=1}^{m-1} q_r g_r. \tag{36}$$

Then

$$E_m + P_B \geq \delta(m-6), \qquad E_m := 2 \sum_{1 \leq i < j \leq m} w(y_j - y_i). \tag{37}$$

Moreover,

$$\boxed{\sum_{r=1}^{m-1} q_r = \beta(m-6)}, \qquad \min_j p_j \leq q_r \leq \beta. \tag{38}$$

Proof. Sum the local inequality over the $m-6$ consecutive seven-point windows. A fixed pair separated by $s \leq 6$ consecutive gaps appears in at most $7-s$ such windows and has coefficient $2/(7-s)$ in each appearance. Its accumulated coefficient is therefore at most 2, which is the coefficient available in E_m. The pressure terms add exactly to P_B, proving (37).

For the mass identity, sum first over windows:

$$\sum_{r=1}^{m-1} q_r = \sum_{a=1}^{m-6} \sum_{j=1}^{6} p_j = \beta(m-6).$$

For a fixed global gap, the relative position j can occur at most once, so $q_r \leq \sum_j p_j = \beta$. Every global gap belongs to at least one of the consecutive seven-point windows, hence $q_r \geq \min_j p_j$. $\square$

Proposition 5.2 (Computer-assisted seven-point inequality). *For the pressure vector* (2)*, every $g_1, \ldots, g_6 \geq 0$ satisfies*

$$\frac{2714(g_1+g_6) + 3733(g_2+g_5) + 3553(g_3+g_4)}{10^7} + \sum_{r=1}^{6} \frac{2}{7-r} \sum_{i=1}^{7-r} w(g_i + \cdots + g_{i+r-1}) \geq \frac{39}{10000}. \tag{39}$$

Computer-assisted proof. The archived verifier performs exhaustive interval branch-and-bound with Arb/FLINT through `python-flint`. The pressure coefficients, target, and grid endpoints are represented by exact rationals. The verifier reconstructs rigorous Arb enclosures for w and its required derivatives from (30)–(31). The grid mesh is 1/4000.

Boxes are discarded only after one of three rigorous lower bounds proves the target: direct interval evaluation, the positive pressure bound, or a tangent bound after positive definiteness of the interval Hessian has been certified. Otherwise the widest coordinate is bisected. Reaching an uncertified terminal grid cell raises an exception; thus a successful run closes the entire search tree.

The finite search used by the archived computation is justified on the unbounded orthant by a pressure cutoff at cell index 57,480 on the mesh 1/4000. Since the smallest pressure coefficient is $2714/10^7$,

$$\frac{2714}{10^7} \frac{57480}{4000} = \frac{1950009}{500000000} = \frac{39}{10000} + \frac{9}{500000000} > \frac{39}{10000},$$

any box with at least one coordinate beyond that cutoff is certified by the pressure term alone. The current default-branch hardening copy of the verifier checks this exact rational identity before constructing the finite branch-and-bound search and aborts if it fails. This additional assertion is not retroactively part of the frozen verifier whose SHA-256 is recorded in Appendix E; the archived run remains the certificate of record, while the v19 assertion makes the domain reduction explicit and fail-closed in the current reproducibility copy.

The frozen 256-bit run reports `verified=true`, with 1,119,372 visited nodes, 559,038 splits, and maximum depth 39. The same frozen verifier also succeeds at 128-bit working precision. The complete 256-bit artifact is publicly archived in the frozen release snapshot [21], identified by its release tag, commit SHA, and cryptographic hashes. Appendix E records the hashes and environment. □

Corollary 5.3 (Coarse span consequence)**.** *For every $m \geq 7$ and $y_1 < \cdots < y_m$,*

$$E_m + \beta(y_m - y_1) \geq \delta(m-6).$$

For the certified vector (2)*,*

$$E_m + \frac{1}{500}(y_m - y_1) \geq \frac{39}{10000}(m-6).$$

Proof. Since $q_r \leq \beta$,

$$P_B = \sum_r q_r g_r \leq \beta \sum_r g_r = \beta(y_m - y_1).$$

Apply Proposition 5.1. □

6 Pressure-preserving block stability

Put

$$\delta := \frac{39}{10000}, \qquad \beta := \frac{1}{500}, \qquad A_m := \delta(m-6), \qquad Q_m := \beta(m-6).$$

For a block $B = (y_1, \ldots, y_m)$, let P_B be the exact pressure (36). Proposition 5.1 gives

$$E_m + P_B \geq A_m, \qquad \sum_{r=1}^{m-1} q_r = Q_m. \tag{40}$$

Remark 6.1 (Immediate accounting gain)**.** *Take $m = 262$. Then*

$$A_{262} = \frac{624}{625} < 1, \qquad Q_{262} = \frac{64}{125}.$$

The proof of the old block-stability lemma applies verbatim with P_B in place of $\beta\,\mathrm{span}(B)$: if $P_B \geq A_{262}$ the claim is trivial, while if $P_B < A_{262}$, the positive lower bound $q_r \geq \min_j p_j$ confines the block span to a fixed compact interval and Lemma 4.2 together with (40) *gives*

$$\mathcal{D}(G_B) + P_B \geq A_{262} - o(1).$$

Averaging the exact pressure as in Section 7 then yields

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \geq \frac{655000 H_{\mathrm{MT}} - 1280}{652504} = 0.6731115225500757511923586\ldots.$$

Thus pressure preservation alone already improves the previous theorem.

6.1 Adjacent-pair pinching

Let $K = (k(y_i - y_j))_{1\le i,j\le m}$ be a limiting Gram matrix. Then $K \succeq 0$, $K_{ii} = 1$, and

$$E_m = \operatorname{tr}(K - I)^2 = 2 \sum_{1\le i<j\le m} w(y_j - y_i). \tag{41}$$

Lemma 6.2 (Adjacent-pair defect). *For consecutive gaps $g_j = y_{j+1} - y_j$,*

$$\boxed{\mathcal{D}(K) \ge \sum_{j=1}^{m-1} w(g_j)}. \tag{42}$$

Proof. Apply two separate pinchings to the original matrix K. The first uses the principal 2×2 blocks

$$(1, 2), (3, 4), \ldots,$$

while the second, independently applied to the same original K, uses

$$(2, 3), (4, 5), \ldots.$$

For each of these two pinchings, the spectral trace functional $X \mapsto \operatorname{tr} \Psi(X)$ cannot increase by the standard pinching/majorization principle [8]: pinching is a random-unitary average and this trace functional is convex and unitarily invariant.

The block associated with a consecutive gap g is

$$\begin{pmatrix} 1 & k(g) \\ \overline{k(g)} & 1 \end{pmatrix},$$

whose eigenvalues are $1 \pm |k(g)| \in [0, 2]$. Its defect is therefore exactly $2|k(g)|^2 = 2w(g)$. The first pinching gives

$$\mathcal{D}(K) \ge 2 \sum_{j \text{ odd}} w(g_j),$$

while the second, as a separate inequality for the same K, gives the corresponding bound over even j. Averaging these two independently obtained inequalities proves (42). $\square$

6.2 A scalar kernel inequality

Set

$$g_0 := \frac{89}{100}, \qquad t := \frac{33}{2}. \tag{43}$$

Lemma 6.3 (Scalar pressure bound). *For every $g \ge 0$ and every $q \in [0, \beta]$,*

$$w(g) + tqg \ge tqg_0. \tag{44}$$

Proof. The normalized kernel has the integral representation

$$k(g) = \int_{-1/2}^{1/2} \frac{\cos(\sqrt{2}\, u)}{\sqrt{2} \sin(1/\sqrt{2})} \cos(2\pi g u)\, du.$$

For $0 \le x \le y \le g_0$ and $|u| \le 1/2$, one has

$$0 \le 2\pi x|u| \le 2\pi y|u| \le \pi g_0 < \pi.$$

Since cosine is decreasing on $[0,\pi]$ and $\cos(\sqrt{2}\,u) > 0$ on $[-1/2, 1/2]$, direct comparison of the two integrands gives

$$k(y) \le k(x).$$

This avoids any differentiation under the integral sign.

The exact-arithmetic verifier accompanying this version certifies the signed bound

$$\frac{171389}{1000000} < k\left(\frac{89}{100}\right). \tag{45}$$

In particular $k(g_0) > 0$, and the preceding monotonicity shows that $k(g) \ge k(g_0) > 0$ for $0 \le g \le g_0$. Squaring therefore preserves the order on this interval. Moreover the same exact-rational verification gives

$$w\left(\frac{89}{100}\right) > \left(\frac{171389}{1000000}\right)^2 > \frac{2937}{100000} = t\beta g_0. \tag{46}$$

If $0 \le g \le g_0$, then

$$w(g) \ge w(g_0) > t\beta g_0 \ge tqg_0.$$

If $g \ge g_0$, then $tqg \ge tqg_0$. This proves (44). $\square$

Lemma 6.4 (Spectral threshold)**.** *Let $K \succeq 0$ be $m \times m$ with diagonal 1, and set*

$$E := \operatorname{tr}(K - I)^2.$$

If $\mathcal{D}(K) < E$, then

$$\mathcal{D}(K) > \frac{m}{m-1}. \tag{47}$$

Proof. If all eigenvalues of K belonged to $[0,2]$, then $\Psi(\lambda) = (\lambda-1)^2$ for every eigenvalue and $\mathcal{D}(K) = E$. Hence $\mathcal{D}(K) < E$ implies the existence of an eigenvalue $1 + a > 2$, with $a > 1$.

The remaining $m-1$ eigenvalues have sum $m-1-a$, hence average $1 - a/(m-1)$. Since Ψ is convex,

$$\mathcal{D}(K) \ge 2a - 1 + (m-1)\Psi\left(1 - \frac{a}{m-1}\right) = 2a - 1 + \frac{a^2}{m-1}.$$

At $a = 1$ the last expression equals $m/(m-1)$, and it is strictly increasing for $a > 1$, proving (47). $\square$

Proposition 6.5 (Strong block stability at $m = 450$)**.** *Let B be a block of*

$$m = 450$$

consecutive retained simple zeros, with exact pressure P_B from (36)*. Let G_B denote the corresponding principal Gram block of M°. Put*

$$A := A_{450} = \frac{4329}{2500}, \qquad Q := Q_{450} = \frac{111}{125}. \tag{48}$$

Uniformly over such blocks,

$$\boxed{\mathcal{D}(G_B) + P_B \ge A - o(1)}. \tag{49}$$

Proof. If $P_B \geq A$, the assertion follows from $\mathcal{D}(G_B) \geq 0$. Assume $P_B < A$. Since $q_r \geq \min_j p_j = 2714/10^7$,

$$\mathrm{span}(B) = \sum_r g_r \leq \frac{P_B}{\min_j p_j} < \frac{A}{\min_j p_j}.$$

Thus all pair separations remain in the fixed compact interval

$$0 \leq |y_i - y_j| \leq R := \frac{A}{\min_j p_j}.$$

Make the uniform error explicit by setting

$$\eta_T(R) := \sup_{\substack{\rho,\rho' \in \mathcal{S}^\circ \\ |x_\rho - x_{\rho'}| \leq R}} \left| \langle v_\rho, v_{\rho'} \rangle - k(x_\rho - x_{\rho'}) \right|.$$

Lemma 4.1 gives $\eta_T(R) \to 0$. Since m is fixed and

$$K_B = (k(y_i - y_j))_{i,j=1}^m,$$

the entrywise bound implies

$$\|G_B - K_B\|_{\mathrm{op}} \leq m\eta_T(R).$$

The function Ψ is globally 2-Lipschitz on $[0, \infty)$; therefore Weyl's eigenvalue perturbation inequality gives the quantitative uniform estimate

$$|\mathcal{D}(G_B) - \mathcal{D}(K_B)| \leq 2m\|G_B - K_B\|_{\mathrm{op}} \leq 2m^2\eta_T(R) = o(1). \tag{50}$$

In particular, the $o(1)$ is uniform over every block in the branch $P_B < A$. It remains to prove

$$\mathcal{D}(K_B) + P_B \geq A.$$

Write $K = K_B$ and $D = \mathcal{D}(K)$. By Proposition 5.1 and (41),

$$E + P_B \geq A, \qquad E = \mathrm{tr}(K - I)^2.$$

Lemma 6.2 and Lemma 6.3 give

$$D + tP_B \geq \sum_{j=1}^{m-1} (w(g_j) + tq_j g_j) \geq tg_0 \sum_{j=1}^{m-1} q_j = tg_0 Q. \tag{51}$$

Suppose for contradiction that $D + P_B < A$. Since $E + P_B \geq A$, we have $D < E$; Lemma 6.4 therefore yields

$$D > \frac{m}{m-1}.$$

Consequently

$$\begin{aligned} D + tP_B &= t(D + P_B) - (t-1)D \\ &< tA - (t-1)\frac{m}{m-1}. \end{aligned} \tag{52}$$

For $m = 450$, the lower bound (51) contradicts (52), because the exact rational margin is

$$g_0 Q + \left(1 - \frac{1}{t}\right)\frac{450}{449} - A = \frac{6363}{30868750} > 0. \tag{53}$$

Hence $D + P_B \geq A$, and the uniform finite-T statement (49) follows. ∎

7 Shifted block pinching with exact pressure

Let

$$x_1 < \cdots < x_{S^\circ}$$

be the normalized ordinates of the retained central set $\mathcal{S}^\circ$. As before,

$$S^\circ = N_0^s(T, 2T) - o(N(T, 2T)).$$

For each of the $m = 450$ offsets modulo m, partition the ordered points into full consecutive blocks of size m, with only $O(m)$ endpoint points left over. Pinching M° to the full principal blocks and the remainder gives

$$\mathcal{D}(M^\circ) \geq \sum_B \mathcal{D}(G_B).$$

Summing over all offsets and applying Proposition 6.5,

$$m\mathcal{D}(M^\circ) \geq (S^\circ - m + 1)A - \sum_B P_B - o(N). \tag{54}$$

Here the accumulated error is genuinely $o(N)$ rather than merely a blockwise notation: across all m offsets there are $O(N)$ full blocks, and (50) bounds the error of each relevant block by $2m^2\eta_T(R)$ with the same $\eta_T(R) \to 0$. Hence the total contribution is $O(N\eta_T(R)) = o(N)$; endpoint blocks contribute only $O(1)$ because m is fixed.

The pressure term retains its exact positional mass. For each fixed relative position r, a given global gap can occupy that position in at most one sliding m-point block. Therefore its coefficient in the sum over all sliding blocks is at most

$$\sum_{r=1}^{m-1} q_r = Q.$$

The span estimate can be made explicit. Because the retained coordinates lie in the sampling interval,

$$x_{S^\circ} - x_1 \leq \frac{LT}{2\pi}.$$

On the other hand Riemann–von Mangoldt gives

$$N(T, 2T) = \frac{T}{2\pi}\big(L + 2\log 2 - 1\big) + O(\log T),$$

and $2\log 2 - 1 > 0$. Thus, for all sufficiently large T, $x_{S^\circ} - x_1 \leq N(T, 2T)$, and therefore

$$\sum_B P_B \leq Q(x_{S^\circ} - x_1) \leq QN(T, 2T). \tag{55}$$

Combining (54) and (55), and absorbing $O(1)$ endpoint terms,

$$\mathcal{D}(M^\circ) \geq \frac{A}{m} N_0^s(T, 2T) - \frac{Q}{m} N(T, 2T) - o(N(T, 2T)). \tag{56}$$

For $m = 450$,

$$\mathcal{D}(M^\circ) \geq \frac{481}{125000} N_0^s(T, 2T) - \frac{37}{18750} N(T, 2T) - o(N(T, 2T)). \tag{57}$$

Proof of Theorem 1.1. Write

$$S = N_0^s(T, 2T), \qquad N = N(T, 2T).$$

Insert (57) into Corollary 3.3:

$$S \geq H_{\mathrm{MT}} N + \frac{481}{125000} S - \frac{37}{18750} N - o(N).$$

Thus

$$\frac{124519}{125000} S \geq \left(H_{\mathrm{MT}} - \frac{37}{18750} \right) N - o(N).$$

Dividing by N and passing to the lower limit gives

$$\begin{aligned}
\liminf_{T\to\infty} \frac{S}{N} &\geq \frac{450 H_{\mathrm{MT}} - 111/125}{450 - 4329/2500} \\
&= \frac{1125000 H_{\mathrm{MT}} - 2220}{1120671} \\
&= 0.6731175265883904388095857434106058666\ldots.
\end{aligned}$$

□

8 Comparison and structural interpretation

At the baseline level, Alpöge–Furman [5] and Lamzouri [7] now provide different unconditional routes to the 67.25%-level Montgomery–Taylor conclusion. The present argument is not obtained by invoking either result: it reconstructs the analytic package used below and then adds the separate stability and local-pressure machinery whose provenance is summarized in the table in Section 1.

Jo's upstream reproducible draft [19] uses $p_j = 1/3000$ and certifies $19/5000 = 0.0038$, leading to $0.6730085279277797\ldots$. Rademacher's subsequent reproducible refinement [20] keeps the same uniform pressure and strengthens the certified local value to $191/50000 = 0.00382$, leading to $0.6730213619501665\ldots$. The position-weighted vector (2) has the same total pressure $1/500$ and the existing computer-assisted certificate proves the stronger local value $39/10000 = 0.0039$.

The previous version of the present paper used the coarse replacement

$$P_B \leq \beta \operatorname{span}(B)$$

inside each block and obtained

$$0.6730732086087052768\ldots.$$

The present version keeps the exact pressure coefficients q_r until the final shifted-block average. With no additional local computation this alone improves the bound to

$$0.6731115225500757512\ldots.$$

The adjacent-pair refinement then gives

$$\boxed{0.6731175265883904388\ldots}.$$

The distinction between the two improvements is structural. Pressure preservation removes an avoidable global accounting loss:

$$\beta(m-1) \quad \longrightarrow \quad \beta(m-6).$$

Adjacent-pair pinching supplies new information about the same block Gram matrix without changing the local seven-point certificate. It is precisely this extra defect lower bound that allows the argument to pass beyond the $A_m < 1$ regime and use $m = 450$.

The natural finite-dimensional local optimization remains

$$\sup_{\substack{p_j \ge 0 \\ \sum p_j = 1/500}} \inf_{g \in \mathbb{R}^6_{\ge 0}} \left(\sum_{j=1}^{6} p_j g_j + \mathcal{E}_7(g) \right). \tag{58}$$

We do not claim that (2) solves this minimax problem, nor that $m = 450$ is optimal for all refinements of the present method. The theorem uses the already certified vector and a new analytic exploitation of the pressure it carries.

9 Scope and trust base

Logical status of the theorem and relation to the Alpöge–Furman/Claude result. The common compression is fixed in Section 2, and the analytic/Gabor package used in the main argument is proved in Appendices A–D. In particular it is not imported from the Montgomery–Taylor clause of Theorem A of Alpöge–Furman [5]. That clause belongs to the unconditional finite-compression/rank–trace framework discovered by Claude and verified and communicated by Alpöge and Furman [5, 6]. In Anthropic's companion Lean artifact, the corresponding optimized result is organized under the historical label Theorem D [9]. Its role here is comparison, provenance, and normalization cross-checking only; no theorem from the Alpöge–Furman paper is a premise of Theorem 1.1.

The analytic external trust base consists of standard classical results: the Weil explicit formula for ζ in the Fourier normalization stated in Appendix C, Stirling's formula, the Prime Number Theorem, the Riemann–von Mangoldt formula and its standard local zero-count consequence, standard Chebyshev and prime-power estimates, the weighted Montgomery–Vaughan Hilbert inequality, Poisson summation, and standard finite-dimensional matrix facts such as von Neumann's trace inequality and the pinching/majorization principle [8]. The Montgomery–Vaughan paper is correctly cited as *Hilbert's inequality*, J. London Math. Soc. (2) **8** (1974), 73–82. The recent pair-correlation papers cited in the introduction provide context; the particular first/second-moment route used here is reconstructed directly in Appendix C.

Proposition 5.2 is computer-assisted. Its trust base includes the frozen artifact release snapshot [21], the frozen verifier, `python-flint`, FLINT/Arb and their dependencies, and the correctness of the execution environment. The 128- and 256-bit runs are replications of the same implementation at different working precisions, not algorithmically independent verifiers.

The refinement from the previous constant to Theorem 1.1 introduces no new seven-point branch-and-bound certificate. Its additional numerical trust base is the small exact-arithmetic verifier `verify_improved_bound.py`. It uses Python integer arithmetic and `Fraction` for all decisive comparisons, together with rigorous rational enclosures for the elementary transcendental quantities. It certifies in particular

$$\frac{171389}{1000000} < k(89/100), \qquad \left(\frac{171389}{1000000}\right)^2 > \frac{2937}{100000},$$

and checks

$$g_0Q + \left(1 - \frac{1}{t}\right)\frac{450}{449} - A = \frac{6363}{30868750} > 0,$$

as well as the final constant. Decimal arithmetic is used only for outward display of certified intervals. This verifier does not replay the existing seven-point branch-and-bound certificate.

The v17 formalization has a separate, explicit trust boundary and a distinct software provenance. The Lean root `HurtadoZeta23.V17FinalAssembly` kernel-checks the implication from exactly two *explicit external certificate propositions*—the archived seven-point claim and the signed kernel claim above—to the published epsilon-form theorem `v17_published_eps_form_of_certificates`. The universal scalar pressure inequality, including the required kernel monotonicity, is proved inside Lean. The external computations are not promoted silently to Lean proof terms or declared as axioms; their propositions remain explicit arguments of the final theorem.

The phrase "exactly two" refers to the explicit mathematical certificate arguments of that final theorem, not to the entire software or foundational stack. The v17 `HurtadoZeta23` modules are compiled as an overlay on the pinned Anthropic `formal-math/zeta23` source tree at commit `fbdc36bbf17d20af3fd0447c6d1a8a02773c9844` [9], together with the pinned Lean/Mathlib toolchain recorded in the repository. Thus the manuscript's logical derivation without importing the Alpöge–Furman theorem as a mathematical premise must not be conflated with clean-room software independence of the Lean formalization.

The pressure vector (2) is an explicit certified improvement, not an optimality theorem. Thus the written mathematical proof is self-contained relative to the classical analytic and matrix results just listed and to the stated Arb/FLINT certificate; it does not claim to reprove those classical theorems from first principles. The separate Lean verification should be read with the software provenance in the preceding paragraph.

A Self-contained derivation of Input I: zero-side inertia

This appendix proves the algebraic finite-compression statement used in Proposition 3.1. The horizontal partner $\rho^\sharp = 1 - \overline{\rho}$ is the composition of conjugation with the functional-equation symmetry; it preserves ordinary ordinate.

A.1 Scope

Let I' be a finite ordinate interval. Since $\rho \mapsto 1 - \overline{\rho}$ preserves ordinary ordinate, the zeros with ordinates in I' form a pairing-invariant finite multiset $Z(I')$ of nontrivial zeta zeros whose ordinates lie in I'. We partition the *distinct* zeros into

$$\mathcal{S}_1 = \{\text{simple critical-line points}\}, \qquad \mathcal{S}_2 = \{\text{multiple critical-line points}\},$$

and a set $\mathcal{P}$ containing one representative from each off-line pair

$$\{\rho, \rho^\sharp\}, \qquad \rho^\sharp := 1 - \overline{\rho}.$$

Write

$$s_1 = \#\mathcal{S}_1, \qquad s_2 = \#\mathcal{S}_2, \qquad p = \#\mathcal{P}.$$

The only zeta-specific fact used in the inertia calculation is the combined functional-equation and conjugation symmetry, which preserves multiplicity:

$$m_{\rho^\sharp} = m_\rho.$$

The finite zero-side Hermitian form obtained by compressing Weil's form to a finite test family can then be written as a pull-back of the block form described below. The present note verifies the resulting inertia statement from scratch.

A.2 Positive index under pull-back

For a Hermitian form H on a finite-dimensional complex vector space, let $n_+(H)$ denote its positive index: the maximal dimension of a subspace on which H is positive definite.

Lemma A.1 (Pull-back monotonicity)**.** *Let H be a Hermitian form on $\mathbb{C}^m$, and let $A : U \to \mathbb{C}^m$ be either complex-linear or conjugate-linear. Then*

$$n_+(H \circ A) \le n_+(H).$$

Proof. Let $U_0 \subset U$ be a complex subspace on which $H \circ A$ is positive definite. Then $A|_{U_0}$ is injective. If A is conjugate-linear, $A(U_0)$ is still a complex subspace, because $\alpha A(u) = A(\overline{\alpha}\, u)$ for every $\alpha \in \mathbb{C}$. Moreover H is positive definite on $A(U_0)$. Hence

$$\dim_{\mathbb{C}} U_0 = \dim_{\mathbb{C}} A(U_0) \le n_+(H).$$

Taking the maximum over U_0 proves the claim. □

Corollary A.2 (Subadditivity)**.** *For Hermitian forms H_1, H_2 on the same space,*

$$n_+(H_1 + H_2) \le n_+(H_1) + n_+(H_2).$$

Proof. Apply Lemma A.1 to $H_1 \oplus H_2$ under the diagonal map $u \mapsto (u, u)$. □

A.3 The off-line hyperbolic block

Lemma A.3 (Signature of one horizontal symmetry pair)**.** *Let $m > 0$. On $\mathbb{C}^2$ define*

$$H_m(x, y) = 2m \,\mathrm{Re}(x\overline{y}).$$

Then

$$H_m(x, y) = \begin{pmatrix} \overline{x} & \overline{y} \end{pmatrix} \begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

and the form has signature $(1, 1)$.

Proof. The Hermitian matrix

$$B_m = \begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix}$$

has characteristic polynomial $\lambda^2 - m^2$, hence eigenvalues m and $-m$. Equivalently, with

$$u = \frac{x + y}{\sqrt{2}}, \qquad v = \frac{x - y}{\sqrt{2}},$$

one has

$$H_m(x, y) = m|u|^2 - m|v|^2.$$

Thus $n_+(H_m) = n_-(H_m) = 1$. □

This is the entire source of the "one positive direction per off-line pair" rule. No limiting argument and no assumption concerning the distance of the pair from the critical line is involved.

A.4 The zero-side block form

For each distinct critical-line zero ρ, let m_ρ be its multiplicity. Introduce one evaluation coordinate $z_\rho \in \mathbb{C}$. For each off-line representative $\rho \in \mathcal{P}$, introduce two coordinates $(z_\rho, z_{\rho^\sharp})$.

Define the Hermitian form

$$\mathcal{Q}(z) = \sum_{\rho \in \mathcal{S}_1 \cup \mathcal{S}_2} m_\rho |z_\rho|^2 + \sum_{\rho \in \mathcal{P}} 2m_\rho \operatorname{Re}\big(z_\rho \overline{z_{\rho^\sharp}}\big). \tag{59}$$

Proposition A.4 (Exact block inertia)**.** *The form $\mathcal{Q}$ is an orthogonal direct sum of $s_1 + s_2$ positive one-dimensional blocks and p hyperbolic two-dimensional blocks. Consequently*

$$n_+(\mathcal{Q}) = s_1 + s_2 + p.$$

If the simple critical-line blocks are removed, the remaining form $\mathcal{Q}'$ satisfies

$$n_+(\mathcal{Q}') = s_2 + p.$$

Proof. Every critical-line block is $m_\rho |z|^2$ with $m_\rho > 0$, and hence has positive index one. Every off-line pair block has signature $(1,1)$ by Lemma A.3. Positive index is additive under orthogonal direct sum, proving both assertions. □

A.5 Compression by evaluation

Let $U = \mathbb{C}^d$ be the coefficient space of the finite family of test functions used to form the compression. For each zero parameter appearing on the zero side, let ℓ_ρ denote the conjugate-linear coefficient functional $c \mapsto \Lambda_\rho(c)$ from Section 2. Define

$$\mathfrak{E}(c) := \big(\ell_\rho(c)\big)_\rho.$$

The map $\mathfrak{E}$ is conjugate-linear in the displayed coefficient convention; Lemma A.1 was formulated to cover this case directly.

By Proposition 2.1, the normalized finite zero-side matrix $\widehat{A}$ is exactly the matrix of the pull-back form

$$c \longmapsto \mathcal{Q}(\mathfrak{E}c).$$

Thus this inertia calculation applies to the same matrix $\widehat{A}$ used in the main argument.

For a simple critical-line zero $\rho \in \mathcal{S}_1$, use the normalized evaluation vector $v_\rho \in \mathbb{C}^d$ defined in Proposition 2.1; thus the normalized evaluation is $c^* v_\rho$. Proposition 2.1 proves for this same family that

$$\|v_\rho\| \le 1.$$

Set

$$P_1 = \sum_{\rho \in \mathcal{S}_1} v_\rho v_\rho^*, \qquad M = V^* V,$$

where V has the v_ρ as columns, and put

$$Q' = \widehat{A} - P_1.$$

Theorem A.5 (Self-contained Input I)**.** *With the preceding notation,*

$$P_1 \succeq 0, \qquad \operatorname{rank} P_1 \le s_1, \qquad \operatorname{tr} P_1 \le s_1,$$

and

$$n_+(Q') \le s_2 + p.$$

In particular, the finite compression cannot acquire more than one positive direction from each multiple critical-line point and each off-line horizontal symmetry pair after the simple critical-line contribution has been separated.

Proof. Each summand $v_\rho v_\rho^*$ is positive semidefinite and rank at most one, so

$$P_1 \succeq 0, \qquad \operatorname{rank} P_1 \le s_1.$$

Moreover

$$\operatorname{tr} P_1 = \sum_{\rho \in \mathcal{S}_1} \|v_\rho\|^2 \le s_1.$$

After deleting the simple critical-line coordinates from the block form $\mathcal{Q}$, the remaining form is $\mathcal{Q}'$ of Proposition A.4, with

$$n_+(\mathcal{Q}') = s_2 + p.$$

The matrix Q' on coefficient space is the pull-back of this form under the corresponding restricted evaluation map. Lemma A.1 therefore gives

$$n_+(Q') \le n_+(\mathcal{Q}') = s_2 + p.$$

□

A.6 Why the functional-equation and conjugation pairing produces the block

For completeness we isolate the algebra behind (59). Weil's Hermitian zero-side form pairs the evaluation at ρ with the evaluation at the horizontal partner under functional-equation plus conjugation symmetry

$$\rho^\sharp = 1 - \overline{\rho}.$$

For an off-line pair of equal multiplicity m_ρ, its two contributions combine as

$$m_\rho \left(\ell_\rho(c)\overline{\ell_{\rho^\sharp}(c)} + \ell_{\rho^\sharp}(c)\overline{\ell_\rho(c)} \right) = 2m_\rho \operatorname{Re}\left(\ell_\rho(c)\overline{\ell_{\rho^\sharp}(c)} \right),$$

which is exactly the hyperbolic block of Lemma A.3. On the critical line, $\rho = \rho^\sharp$, so the same expression collapses to a positive square (up to the normalization convention already absorbed into the zero-side form).

Thus the signature claim is a direct algebraic consequence of functional equation symmetry once the zero-side Hermitian form has been written in evaluation coordinates.

A.7 Adversarial checks

Several possible pathologies do not change the result.

- **Near-line pairs.** As $\operatorname{Re}\rho \to 1/2$, the two evaluation functionals may become nearly dependent, but pull-back can only *decrease* positive index. No uniform linear independence is required.
- **Deep off-line pairs.** The eigenvalues $\pm m_\rho$ of the abstract pair block are independent of the depth $|\operatorname{Re}\rho - 1/2|$. The evaluation map may distort or annihilate directions, but cannot increase n_+.
- **Coincident ordinates.** Different zero points may have the same ordinate. This creates dependence among evaluation functionals, which again can only lower the pull-back positive index.
- **Multiplicity.** A multiple critical-line point contributes one positive coordinate block $m_\rho |z|^2$, hence one positive direction, regardless of m_ρ. Multiplicity changes the weight but not its positive index.
- **Noninjective compression.** No injectivity of the full evaluation map is assumed; this is precisely why Lemma A.1 is formulated for arbitrary linear maps.

B Self-contained derivation of Input II: exterior-zero tail

This appendix proves the trace-class tail estimate and its effect on the rank–trace functional. The exterior matrix is grouped into Hermitian critical-line squares and off-line pair terms $uv^* + vu^*$, so that $G = A + E$ holds entrywise under the same zero-side convention.

B.1 Setup

Let

$$\ell = \log \frac{T}{2\pi}, \qquad L = \ell, \qquad X = e^L = \frac{T}{2\pi}, \qquad D_0 = T^{1/2},$$

and let

$$I = (T, 2T], \qquad I' = (T - D_0,\ 2T + D_0].$$

Let the critical sampling grid be

$$\tau_k = T + \frac{2\pi k}{L}, \qquad 0 \le k < d, \qquad d = \left\lfloor \frac{LT}{2\pi} \right\rfloor.$$

The tapered Montgomery–Taylor window ϕ has support in $[-L/2, L/2]$, satisfies $0 \le \phi \le 1$, and for fixed taper width has

$$C_1 := \|\phi''\|_1 = O(1).$$

Put

$$a = \frac{1}{L} \int_{\mathbb{R}} \phi(u)^2 \, du.$$

For this window, $a \to K(0) > 0$; hence $a \asymp 1$.

Use the globally fixed test family (4). Thus

$$G_{jk} = W(f_j, f_k)$$

is exactly the common matrix (8). Using the zero-side expansion of the normalized Weil formula, define A_{jk} by restricting that zero sum to zeros whose ordinary ordinates lie in I', grouping every off-line zero together with its horizontal partner $\rho^\sharp = 1 - \overline{\rho}$. Define E_{jk} by the complementary exterior zero sum. Since the inside and outside sets form a disjoint partition invariant under $\rho \mapsto \rho^\sharp$, one has entrywise

$$\boxed{G = A + E.}$$

Equivalently,

$$\widehat{A} = \widehat{G} - \widehat{E}, \qquad \widehat{G} = \frac{G}{aL^2}, \quad \widehat{A} = \frac{A}{aL^2}, \quad \widehat{E} = \frac{E}{aL^2}.$$

B.2 Off-axis Fourier decay

We use the standard Paley–Wiener integration-by-parts bound.

Lemma B.1 (Complex-shift decay)**.** *Let $\phi \in C_c^2([-L/2, L/2])$. For real $r \neq 0$ and $|y| < 1/2$,*

$$|\widehat{\phi}(r - iy)| \leq e^{L|y|/2}\, \|\phi''\|_1\, |r - iy|^{-2} \leq X^{1/4} C_1\, |r|^{-2}.$$

Proof. Twice integrating by parts in

$$\widehat{\phi}(r - iy) = \int_{-L/2}^{L/2} \phi(u) e^{-iru - yu}\, du$$

and using the compactly supported C^2 taper gives

$$\widehat{\phi}(r - iy) = -(r - iy)^{-2} \int \phi''(u) e^{-i(r-iy)u}\, du.$$

Since $|e^{-yu}| \leq e^{L|y|/2}$ on the support,

$$|\widehat{\phi}(r - iy)| \leq e^{L|y|/2} C_1 |r - iy|^{-2}.$$

Now $|r - iy| \geq |r|$ and, because $|y| < 1/2$,

$$e^{L|y|/2} \leq e^{L/4} = X^{1/4}.$$

□

B.3 One external zero

Let $\rho = \beta + i\gamma$ be a distinct zero with $\gamma \notin I'$ and put

$$y = \beta - \frac{1}{2}, \qquad |y| < \frac{1}{2}.$$

Define its sampled vector

$$u_\rho = \left(\widehat{\phi}((\gamma - \tau_k) - iy)\right)_{0 \leq k < d}.$$

Let

$$D = \operatorname{dist}(\gamma, I) \geq D_0.$$

Lemma B.2 (Tail vector bound)**.** *Uniformly for every such zero,*

$$\|u_\rho\|_2^2 \ll C_1^2 X^{1/2} L D^{-3}.$$

Proof. Lemma B.1 gives
$$\|u_\rho\|_2^2 \le C_1^2 X^{1/2} \sum_{0\le k<d} |\gamma-\tau_k|^{-4}.$$

The grid spacing is $h = 2\pi/L$. Comparing the monotone sum with an integral,

$$\sum_{0\le k<d} |\gamma-\tau_k|^{-4} \ll D^{-4} + \frac{1}{h}\int_D^\infty t^{-4}\,dt \ll D^{-4} + LD^{-3} \ll LD^{-3},$$

because $D \ge 1$ and $L \ge 2$ for large T. □

B.4 Summing over zeros outside the enlarged interval

We use only the classical estimate

$$N(t+1) - N(t) \le A_0 \log(t+3) \tag{60}$$

for some absolute A_0 and all $t \ge 0$.

Lemma B.3 (Weighted exterior zero count)**.** *Counting zeros with multiplicity,*

$$\sum_{\gamma\notin I'} m_\rho \ \mathrm{dist}(\gamma, I)^{-3} \ll \frac{\log T}{D_0^2}.$$

Proof. For the upper tail, group zeros into intervals

$$2T + D_0 + j < \gamma \le 2T + D_0 + j + 1, \qquad j \ge 0.$$

By (60), the contribution is
$$\ll \sum_{j\ge 0} \frac{\log(2T + D_0 + j + 4)}{(D_0 + j)^3}.$$

Splitting at $j = T$ gives
$$\ll \frac{\log T}{D_0^2}.$$

The lower positive tail $0 < \gamma < T - D_0$ is identical. Zeros with $\gamma \le 0$ are at distance $\gg T$ from I and contribute $O(T^{-2}\log T)$, which is smaller. Adding the three ranges proves the claim. □

B.5 Trace-norm tail bound

The exterior zero-side matrix must be grouped according to the horizontal symmetry involution. If u_ρ is the sampled evaluation column, then

$$E = \sum_{\substack{\rho \text{ exterior} \\ \rho=\rho^\sharp}} m_\rho u_\rho u_\rho^* + \sum_{\substack{\{\rho,\rho^\sharp\} \text{ exterior} \\ \rho\ne\rho^\sharp}} m_\rho \left(u_\rho u_{\rho^\sharp}^* + u_{\rho^\sharp} u_\rho^* \right).$$

This is Hermitian term-by-term after pairing. Moreover

$$\|uv^* + vu^*\|_1 \le 2\|u\|_2\|v\|_2 \le \|u\|_2^2 + \|v\|_2^2.$$

Hence the scalar tail-vector estimate may be summed exactly as before.

Proposition B.4 (Independent tail estimate)**.** *One has*

$$\|E\|_1 \ll C_1^2 X^{1/2} L \frac{\log T}{D_0^2}.$$

Consequently

$$\|\widehat{E}\|_1 = \frac{\|E\|_1}{aL^2} \ll \frac{X^{1/2}\log T}{LD_0^2}.$$

At $\lambda = 1$, $L = \ell$, $X = T/(2\pi)$ and $D_0 = T^{1/2}$, hence

$$\boxed{\|\widehat{E}\|_1 \ll T^{-1/2}}.$$

Proof. For critical-line exterior zeros use $\|u_\rho u_\rho^*\|_1 = \|u_\rho\|_2^2$. For each off-line pair use the preceding inequality. Both members have the same ordinate and opposite horizontal displacement, so Lemma B.2 applies to each. Summing with multiplicity and using Lemma B.3 gives the stated unnormalised bound. Division by aL^2, with $a \asymp 1$, gives the normalized estimate. □

B.6 The rank–trace functional under a trace-class perturbation

Define

$$\mathcal{R}(H) = 4\operatorname{tr} H - \|H\|_{\mathrm{F}}^2$$

for a finite Hermitian matrix H.

Lemma B.5 (Perturbation inequality)**.** *If $A = G - E$ are Hermitian matrices and $\varepsilon = \|E\|_1$, then*

$$\mathcal{R}(A) \geq \mathcal{R}(G) - 4\varepsilon - 2\|G\|_{\mathrm{F}}\varepsilon - \varepsilon^2. \tag{61}$$

Proof. Since $|\operatorname{tr} E| \leq \|E\|_1 = \varepsilon$,

$$4\operatorname{tr} A \geq 4\operatorname{tr} G - 4\varepsilon.$$

Also

$$\|A\|_{\mathrm{F}} \leq \|G\|_{\mathrm{F}} + \|E\|_{\mathrm{F}} \leq \|G\|_{\mathrm{F}} + \|E\|_1 = \|G\|_{\mathrm{F}} + \varepsilon.$$

Squaring and combining the two inequalities proves (61). □

Theorem B.6 (Self-contained Input II)**.** *For the optimized bandwidth $\lambda = 1$, the prime-side moment estimate of Appendix C gives $\|\widehat{G}\|_{\mathrm{F}} = O(N(T, 2T)^{1/2})$. Then*

$$\boxed{4\operatorname{tr}\widehat{A} - \|\widehat{A}\|_{\mathrm{F}}^2 \geq 4\operatorname{tr}\widehat{G} - \|\widehat{G}\|_{\mathrm{F}}^2 - o(N(T, 2T)).}$$

Proof. Apply Lemma B.5 to the normalized matrices with

$$\varepsilon = \|\widehat{E}\|_1 \ll T^{-1/2}$$

from Proposition B.4. Since $N(T, 2T) \asymp T\log T$,

$$\varepsilon = o(N), \qquad \varepsilon^2 = o(N),$$

and using the Appendix C bound,

$$\|\widehat{G}\|_{\mathrm{F}}\varepsilon = O(N^{1/2}T^{-1/2}) = o(N).$$

Substitution in (61) proves the result. □

Remark B.7 (Package closure)**.** *Appendix C establishes the prime-side second moment without using the present tail comparison. Thus using $\|\widehat{G}\|_{\mathrm{F}} = O(N^{1/2})$ here is logically acyclic.*

C Self-contained derivation of Input III: prime-side moments

This appendix fixes the Fourier normalization of Weil's explicit formula [1, 4], derives the spectral density, controls integrated finite-grid end effects, and evaluates the prime diagonal and both off-diagonal frequency types. The weighted Montgomery–Vaughan inequality [3] is used only with an unspecified absolute constant. The route is deliberately written out from these classical ingredients. Its prime-side architecture parallels Section 5 of Alpöge–Furman [5]—continuous second moment, diagonal prime term, quotient-frequency off-diagonal terms controlled by Montgomery–Vaughan, and secondary terms—but no theorem from that paper is invoked as a premise here. Thus "self-contained" in this appendix is a statement about logical derivation from the cited classical inputs, not a claim of independent discovery of the base analytic architecture.

C.1 Classical input and notation

Put

$$\ell = \log\frac{T}{2\pi}, \qquad L = \ell, \qquad X = e^L = \frac{T}{2\pi}, \qquad I = [T, 2T],$$

and define

$$\ell_1 := \ell + 2\log 2 - 1. \tag{62}$$

Thus the Riemann–von Mangoldt formula reads

$$N(T, 2T) = \frac{T\ell_1}{2\pi} + O(\ell).$$

Let $\phi = \phi_L \in C_c^2(\mathbb{R})$ be real, even, supported exactly on $[-L/2, L/2]$, with a fixed-width endpoint taper and uniform derivative bounds. Define

$$\widehat{\phi}(r) = \int_{\mathbb{R}} \phi(u) e^{-iru}\,du, \qquad \Phi(r) = \int_{\mathbb{R}} \phi(u)^2 e^{-iru}\,du,$$

$$a = \frac{1}{L}\int \phi^2, \qquad b = \frac{1}{L}\int \phi^4, \qquad g = \phi^2 * \phi^2.$$

For the optimized family, a and b remain bounded above and below by positive constants.

We use the classical Weil explicit formula in the following fixed normalization. With

$$\widehat{q}(z) = \int_{\mathbb{R}} q(u) e^{izu}\,du, \qquad q(u) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{q}(t) e^{-itu}\,dt,$$

the external theorem used here is

$$\begin{aligned}\sum_{\rho} m_\rho \widehat{q}(z_\rho) = \widehat{q}(i/2) + \widehat{q}(-i/2) - \sum_{n\geq 1}\frac{\Lambda(n)}{\sqrt{n}}\left(q(\log n) + q(-\log n)\right)\\ + \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{q}(\tau)\left[\Re\frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{i\tau}{2}\right) - \log\pi\right] d\tau.\end{aligned} \tag{63}$$

Here $z_\rho = -i(\rho - \frac{1}{2})$ and zeros are counted with multiplicity. This is the standard Guinand–Weil explicit formula in the present Fourier normalization; see Weil [1] and, for a modern formulation with explicit strip hypotheses, Iwaniec–Kowalski [2, Theorem 5.12]. We apply it to $q = f * \widetilde{g}$, $\widetilde{g}(u) = \overline{g(-u)}$. Since $f, g \in C_c^2([-L/2, L/2])$, one has $q \in C_c^2([-L, L])$. Moreover

$$\widehat{q}(z) = \mathcal{F}_+ f(z)\,\overline{\mathcal{F}_+ g(\overline{z})},$$

and two integrations by parts in each factor give, uniformly on every fixed strip $|\Im z| \le \frac{1}{2} + \varepsilon$,

$$|\widehat{q}(z)| \ll_{\varepsilon,L,f,g} (1 + |z|)^{-4}.$$

Thus the standard strip-decay hypotheses of the explicit formula are satisfied. The support condition also implies that only $n \le X = e^L$ occur in the prime sum. Fourier inversion then gives the following proposition.

Proposition C.1 (Normalized Weil spectral density)**.** *The compressed Weil form has density*

$$\nu_X(\tau) = \mu(\tau) + \Pi_X(\tau) + P_X(\tau),$$

where

$$\mu(\tau) = \frac{1}{2\pi}\Re\frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{i\tau}{2}\right) - \frac{\log\pi}{2\pi}, \tag{64}$$

$$P_X(\tau) = -\frac{1}{\pi}\sum_{n\le X}\frac{\Lambda(n)}{\sqrt{n}}\cos(\tau\log n), \tag{65}$$

$$\Pi_X(\tau) = \frac{1}{2\pi(1/4+\tau^2)} + \frac{1}{\pi}\Re\frac{X^{1/2+i\tau}-1}{1/2+i\tau}. \tag{66}$$

More precisely,

$$W(f,g) = \int_{\mathbb{R}} \mathcal{F}_+f(\tau)\overline{\mathcal{F}_+g(\tau)}\nu_X(\tau)\,d\tau.$$

Proof. The gamma contribution in the classical explicit formula is the integral against μ. For the prime term,

$$q(y) + q(-y) = \frac{1}{\pi}\int_{\mathbb{R}} \mathcal{F}_+f(\tau)\overline{\mathcal{F}_+g(\tau)}\cos(\tau y)\,d\tau,$$

and $q(\pm\log n) = 0$ for $n > X$. For the pole terms,

$$H(i/2) + H(-i/2) = \int_{\mathbb{R}} q(u)(e^{-u/2} + e^{u/2})\,du.$$

Since $\operatorname{supp} q \subset [-L, L]$, this equals

$$\int_{\mathbb{R}} q(u)e^{-|u|/2}\,du + \int_{-L}^{L} q(u)e^{|u|/2}\,du.$$

With $s = 1/2 + i\tau$,

$$\int_{\mathbb{R}} e^{-|u|/2}e^{-i\tau u}du = (1/4+\tau^2)^{-1},$$

while

$$\int_{-L}^{L} e^{|u|/2}e^{-i\tau u}du = 2\Re\frac{X^{\bar{s}}-1}{\bar{s}} = 2\Re\frac{X^s-1}{s}.$$

Fourier inversion therefore gives exactly the density Π_X. This proves the spectral identity with all signs and 2π factors fixed. $\square$

Stirling gives, uniformly for $\tau \in I$,

$$\mu(\tau) = \frac{1}{2\pi}\log\frac{\tau}{2\pi} + O(T^{-2}), \qquad \mu'(\tau) = O(T^{-1}).$$

Also

$$|P_X(\tau)| \ll \sqrt{X}, \qquad |\Pi_X(\tau)| \ll \frac{\sqrt{X}}{1+|\tau|},$$

and Riemann–von Mangoldt yields

$$\int_T^{2T} \mu(\tau) d\tau = N(T, 2T) + O(\ell).$$

Let

$$h = \frac{2\pi}{L}, \qquad d = \left\lfloor \frac{T}{h} \right\rfloor, \qquad \tau_k = T + kh \quad (0 \le k < d).$$

By (9), the same common matrix G satisfies

$$G_{jk} = \int_{\mathbb{R}} \widehat{\phi}(\tau - \tau_j)\widehat{\phi}(\tau - \tau_k)\nu_X(\tau)\, d\tau,$$

and set

$$\widetilde{G} := \frac{G}{L}, \qquad \widehat{G} := \frac{\widetilde{G}}{aL} = \frac{G}{aL^2}.$$

This distinction between G, $\widetilde{G}$ and $\widehat{G}$ is maintained throughout.

C.2 Fourier and prime-power estimates

The fixed-width C^2 taper gives

$$\max\{|\widehat{\phi}(r)|, |\Phi(r)|\} \le \psi(r) := \min\left\{L, \frac{C_1}{|r|}, \frac{C_2}{r^2}\right\}, \tag{67}$$

with constants independent of L. Consequently

$$\Psi_0 := \int_0^\infty \psi(r)\, dr \ll \log L, \qquad \int_{\mathbb{R}} \psi(r)^2|r|\, dr \ll \log L, \qquad \int_{\mathbb{R}} \psi(r)^2\, dr \ll L. \tag{68}$$

Plancherel and Fourier inversion give

$$\int_{\mathbb{R}} \Phi(x)^2\, dx = 2\pi bL, \qquad \int_{\mathbb{R}} \Phi(x)^2 e^{ixy}\, dx = 2\pi g(y). \tag{69}$$

We also use the classical prime-power estimates

$$\sum_{n\le x} \frac{\Lambda(n)}{\sqrt{n}} \ll \sqrt{x}, \qquad \sum_{n\le x} \Lambda(n)^2 \ll x\log x, \tag{70}$$

and

$$\sum_{n\le x} \frac{\Lambda(n)^2}{n} = \frac{1}{2}\log^2 x + O(\log x). \tag{71}$$

The asymptotic (71) follows from the Prime Number Theorem by standard partial summation; the upper bounds in (70) require only Chebyshev-strength estimates. We use Titchmarsh [4] as a standard reference.

C.3 First trace moment

Put

$$S_d(\tau) := \sum_{0\le k<d} |\widehat{\phi}(\tau-\tau_k)|^2, \qquad S_\infty(\tau) := \sum_{k\in\mathbb{Z}} |\widehat{\phi}(\tau-\tau_k)|^2.$$

For the infinite translated critical grid, Poisson summation gives

$$S_\infty(\tau) = aL^2 \qquad (\tau\in\mathbb{R}). \tag{72}$$

We require two distinct end-effect estimates. The first controls the lattice points omitted from the finite grid while τ remains in I:

$$R_{\rm in} := \int_I (S_\infty(\tau) - S_d(\tau))|\nu_X(\tau)|\, d\tau.$$

On I one has

$$|\nu_X(\tau)| \ll B, \qquad B := \ell + \sqrt{X},$$

by Stirling and the displayed bounds for P_X, Π_X. Using (67), the left omitted tail is bounded by

$$\sum_{j\ge1} \int_{jh}^\infty \psi(r)^2\, dr,$$

and the right omitted tail is identical up to $O(1)$ endpoint indices. Hence, by Tonelli and (68),

$$R_{\rm in} \ll B\left(\frac{1}{h}\int_0^\infty r\psi(r)^2\, dr + \int_0^\infty \psi(r)^2\, dr\right) \ll BL\log L. \tag{73}$$

The second estimate, absent from a naive truncation to I, controls the actual finite-grid mass outside I:

$$R_{\rm out} := \int_{\mathbb{R}\setminus I} S_d(\tau)|\nu_X(\tau)|\, d\tau.$$

For the prime and pole pieces the global bounds

$$|P_X(\tau)| \ll \sqrt{X}, \qquad |\Pi_X(\tau)| \ll 1+\sqrt{X}$$

give, by the same endpoint-tail summation,

$$\int_{\mathbb{R}\setminus I} S_d(\tau)(|P_X(\tau)| + |\Pi_X(\tau)|)\, d\tau \ll \sqrt{X}\, L\log L. \tag{74}$$

For the archimedean term, Stirling gives $|\mu(\tau)| \ll \ell$ on $|\tau|\le 3T$. Thus the part of $\mathbb{R}\setminus I$ with $|\tau|\le 3T$ is $O(\ell L\log L)$ by the same argument. If $|\tau|>3T$, then every $\tau_k\in[T, 2T+O(h)]$ satisfies $|\tau-\tau_k| \asymp |\tau|$, and (67) gives

$$S_d(\tau) \ll d\,|\tau|^{-4} \ll TL\,|\tau|^{-4}, \qquad |\mu(\tau)| \ll \log(2+|\tau|).$$

Consequently

$$\int_{|\tau|>3T} S_d(\tau)|\mu(\tau)|\, d\tau \ll TL\int_{3T}^\infty \frac{\log(2+t)}{t^4}\, dt \ll \frac{L\ell}{T^2}.$$

Combining these estimates,

$$R_{\rm out} \ll (\ell+\sqrt{X})L\log L + \frac{L\ell}{T^2} = o(L^2N) \tag{75}$$

for $L \asymp \ell$, $X \asymp T$, since $N \asymp T\ell$.

Now the spectral representation gives the exact decomposition

$$\operatorname{tr} G = \int_I S_d(\tau)\nu_X(\tau)\,d\tau + \int_{\mathbb{R}\setminus I} S_d(\tau)\nu_X(\tau)\,d\tau.$$

Using (72), (73), and (75),

$$\operatorname{tr} G = aL^2 \int_I \nu_X(\tau)\,d\tau + o(L^2 N). \tag{76}$$

By Riemann–von Mangoldt and Stirling,

$$\int_I \mu(\tau)\,d\tau = N + O(\ell).$$

Termwise integration and (70) give

$$\left|\int_I P_X(\tau)\,d\tau\right| \ll \sum_{n\le X} \frac{\Lambda(n)}{\sqrt{n}\log n} \ll \sqrt{X} = o(N),$$

while the explicit formula for Π_X gives $\int_I \Pi_X(\tau)\,d\tau = o(N)$. Therefore

$$\operatorname{tr} \widetilde{G} = aLN(1+o(1)), \qquad \boxed{\operatorname{tr} \widehat{G} = N(1+o(1)).} \tag{77}$$

C.4 Integrated finite-grid end effects for the second moment

Define

$$K(\tau,\tau') = \sum_{0\le k<d} \widehat{\phi}(\tau-\tau_k)\widehat{\phi}(\tau'-\tau_k),$$

$$K_\infty(\tau,\tau') = \sum_{k\in\mathbb{Z}} \widehat{\phi}(\tau-\tau_k)\widehat{\phi}(\tau'-\tau_k) = L\Phi(\tau-\tau'),$$

and $K_{\text{out}} = K_\infty - K$. Since $\sum_{k\in\mathbb{Z}} |\widehat{\phi}(\tau-\tau_k)|^2 = aL^2$, Cauchy–Schwarz gives

$$|K|, |K_\infty| \le aL^2, \qquad |K_{\text{out}}(\tau,\tau')| \le \sum_{k\notin[0,d)} \psi(\tau-\tau_k)\psi(\tau'-\tau_k). \tag{78}$$

Put

$$B = C(\ell+\sqrt{X}),$$

so $|\nu_X(\tau)| \le B$ for $|\tau| \le 4T$ and $B^2 \ll \ell^2 + X$.

Lemma C.2 (Global density bound used in the tails)**.** *For $|\tau| \le 4T$,*

$$|\nu_X(\tau)| \ll \ell + \sqrt{X} =: B.$$

If $\tau_k \in [T,2T]$ and $r = |\tau-\tau_k| > 2T$, then

$$|\nu_X(\tau)| \ll B + \log(r/T).$$

Proof. The prime term satisfies $|P_X| \ll \sqrt{X}$ by $\sum_{n\le X} \Lambda(n)/\sqrt{n} \ll \sqrt{X}$. The pole density satisfies $|\Pi_X(\tau)| \ll \sqrt{X}/(1+|\tau|) + 1/(1+\tau^2)$. Stirling gives $\mu(\tau) \ll 1 + \log(2+|\tau|)$. This proves the first estimate on $|\tau| \le 4T$. For $r > 2T$ and $\tau_k \in [T,2T]$, one has $|\tau| \le r + 2T < 2r$ and hence $\log(2+|\tau|) \ll \ell + \log(r/T)$; the prime term remains bounded by $O(\sqrt{X})$ and the pole term is smaller. □

Proposition C.3 (Integrated end effects)**.** *For $I = (T, 2T]$ and*

$$M = \iint_{I\times I} \Phi(\tau-\tau')^2 \nu_X(\tau)\nu_X(\tau')\, d\tau d\tau',$$

one has

$$\operatorname{tr}\widetilde{G}^2 = M + O\Big(L\ell \log L\, (\ell^2 + X)\Big), \qquad \widetilde{G} = G/L. \tag{79}$$

Proof. Expanding $\operatorname{tr} G^2$ gives

$$L^2 \operatorname{tr}\widetilde{G}^2 = \iint_{\mathbb{R}^2} |K(\tau,\tau')|^2 \nu_X(\tau)\nu_X(\tau')\, d\tau d\tau'.$$

On $I\times I$, $|K_\infty|^2 = L^2\Phi^2$. Write the difference from $L^2 M$ as $E_1 + E_2$, where

$$E_1 = \iint_{I\times I} (|K|^2 - |K_\infty|^2)\nu\nu', \qquad E_2 = \iint_{(I\times I)^c} |K|^2 \nu\nu'.$$

For E_1,

$$\big||K|^2 - |K_\infty|^2\big| \le |K-K_\infty|(|K| + |K_\infty|) = |K_{\text{out}}|(|K|+|K_\infty|) \ll L^2 |K_{\text{out}}|.$$

Hence

$$|E_1| \ll L^2 B^2 \sum_{k\notin[0,d)} \left(\int_I \psi(\tau-\tau_k)\, d\tau\right)^2.$$

For omitted grid points at distance $\Delta > 0$ from I,

$$\int_I \psi(\tau-\tau_k) d\tau \le \int_\Delta^\infty \psi(r) dr \le \min\left\{\Psi_0, \frac{C}{\Delta}\right\}.$$

At the left edge, omitted points are $k=-j$, $j\ge 1$, and have distance jh from I. At the right edge the two exceptional omitted indices $k = d, d+1$ may lie in, or within one mesh step of, the endpoint $2T$ because $d = \lfloor T/h \rfloor$; for them we use the crude bound $2\Psi_0$. Every further right-hand omitted index has distance at least jh, $j \ge 1$. Therefore

$$\begin{aligned} S &:= \sum_{k\notin[0,d)} \left(\int_I \psi(\tau-\tau_k) d\tau\right)^2 \\ &\ll \Psi_0^2 + \sum_{j\ge1} \min\left\{\Psi_0, \frac{C}{jh}\right\}^2. \end{aligned}$$

Let $J = \lceil C/(h\Psi_0) \rceil$. Then

$$\sum_{j\le J} \Psi_0^2 \ll \frac{\Psi_0}{h} + \Psi_0^2,$$

whereas

$$\sum_{j>J} \frac{C^2}{j^2h^2} \ll \frac{C^2}{h^2 J} \ll \frac{\Psi_0}{h}.$$

Since $h = 2\pi/L$ and $\Psi_0 = O(\log L)$,

$$S = O(L\log L).$$

Thus

$$|E_1| \ll L^3 B^2 \log L. \tag{80}$$

For E_2, symmetry lets us place the first variable outside I and multiply by two. Since $|K|^2 \ll L^2|K|$ and $|K| \le \sum_{0\le k<d} \psi(\tau-\tau_k)\psi(\tau'-\tau_k)$,

$$|E_2| \ll L^2 \sum_{0\le k<d} A_k C_k,$$

where

$$A_k = \int_{\tau\notin I} \psi(\tau-\tau_k)|\nu_X(\tau)|\,d\tau, \quad C_k = \int_{\mathbb{R}} \psi(\tau'-\tau_k)|\nu_X(\tau')|\,d\tau'.$$

We first bound C_k uniformly. On $|\tau'-\tau_k| \le 2T$ one has $|\tau'| \le 4T$, hence

$$\int_{|\tau'-\tau_k|\le 2T} \psi(\tau'-\tau_k)|\nu_X(\tau')|d\tau' \le 2B\Psi_0.$$

For $r = |\tau'-\tau_k| > 2T$, one has $|\tau'| \le 2r$ and

$$|\nu_X(\tau')| \ll B + \log(r/T), \qquad \psi(r) \ll r^{-2}.$$

Therefore the far part is

$$\ll \int_{2T}^{\infty} \frac{B+\log(r/T)}{r^2}\,dr \ll \frac{B+1}{T} \ll B,$$

and hence

$$C_k \ll B\Psi_0. \tag{81}$$

Now sum A_k before integrating. Put

$$\sigma(\tau) = \sum_{0\le k<d} \psi(\tau-\tau_k), \qquad \Delta = \operatorname{dist}(\tau, I).$$

The distances to grid points dominate $\Delta, \Delta+h, \Delta+2h, \ldots$, so monotonicity yields

$$\sigma(\tau) \le \psi(\Delta) + \frac{1}{h}\int_{\Delta}^{\infty} \psi(r)dr \ll \psi(\Delta) + L\min\left\{\Psi_0, \frac{C}{\Delta}\right\}. \tag{82}$$

Also, since $d \ll LT$ and $\psi(r) \ll r^{-2}$,

$$\sigma(\tau) \ll LT\,\Delta^{-2}. \tag{83}$$

For $0 < \Delta \le 2T$, $|\tau| \le 4T$, so $|\nu_X| \le B$. Integrating (82) on the two sides of I gives

$$\begin{aligned}\int_{0<\Delta\le 2T} |\nu_X(\tau)|\sigma(\tau)d\tau &\ll B\left[\Psi_0 + L\int_0^{2T} \min\left\{\Psi_0, \frac{C}{\Delta}\right\} d\Delta\right] \\ &\ll B\left[\Psi_0 + L + L\log(2T\Psi_0/C)\right] \\ &\ll BL\ell.\end{aligned}$$

For $\Delta > 2T$, use (83) and $|\nu_X(\tau)| \ll B + \log(\Delta/T)$:

$$\begin{aligned}\int_{\Delta>2T} |\nu_X(\tau)|\sigma(\tau)d\tau &\ll LT\int_{2T}^{\infty} \frac{B+\log(\Delta/T)}{\Delta^2} d\Delta \\ &\ll LB.\end{aligned}$$

Consequently

$$\sum_k A_k = \int_{\tau \notin I} |\nu_X(\tau)| \sigma(\tau) d\tau \ll BL\ell. \tag{84}$$

Combining (81) and (84),

$$|E_2| \ll L^2 (B\Psi_0)(BL\ell) \ll L^3 B^2 \ell \log L. \tag{85}$$

Equations (80) and (85) give

$$L^2 \operatorname{tr} \widetilde{G}^2 = L^2 M + O(L^3 B^2 \ell \log L).$$

Divide by L^2 and use $B^2 \ll \ell^2 + X$ to obtain (79). $\square$

At $L = \ell$ and $X \asymp T$, the error in (79) is $o(TL\ell^2)$.

C.5 Evaluation of the continuous second moment

For functions u_1, u_2 on I, write

$$\mathcal{M}[u_1, u_2] = \iint_{I \times I} \Phi(\tau - \tau')^2 u_1(\tau) u_2(\tau')\, d\tau\, d\tau'.$$

Then

$$M = \mathcal{M}[\mu, \mu] + \mathcal{M}[P_X, P_X] + 2\mathcal{M}[\mu, P_X] + 2\mathcal{M}[\mu, \Pi_X] + 2\mathcal{M}[P_X, \Pi_X] + \mathcal{M}[\Pi_X, \Pi_X].$$

C.5.1 Archimedean term

Using $\mu(\tau) = \mu(\tau') + O(|\tau - \tau'|/T)$, (68), and (69), one obtains

$$\mathcal{M}[\mu, \mu] = 2\pi b L \int_I \mu(\tau)^2\, d\tau + O(\ell^2 \log L) = \frac{bLT\ell_1^2}{2\pi} + o(TL\ell^2). \tag{86}$$

Indeed, the local replacement error is bounded by $O(\ell \int \Phi(x)^2 |x| dx)$ after integrating over I, and the truncation of the full x-integral at the two interval ends costs $O(\ell^2 \int \Phi(x)^2 \min(|x|, T) dx)$.

C.5.2 Prime term: diagonal and two off-diagonal families

Write

$$a_n = \frac{\Lambda(n)}{\sqrt{n}}, \qquad y_n = \log n,$$

so

$$P_X(\tau) = -\frac{1}{2\pi} \sum_{n \le X} a_n (e^{i\tau y_n} + e^{-i\tau y_n}).$$

Substitute $x = \tau - \tau'$. For $|x| < T$, the τ'-interval is $[T + x_-, 2T - x_+]$, $x_\pm = \max(\pm x, 0)$. Expanding the four exponential products splits $\mathcal{M}[P_X, P_X]$ into a quotient-frequency diagonal D, a quotient-frequency off-diagonal O_1, and a product-frequency term O_2.

For $n = m$, the inner interval has length $T - |x|$, hence Fourier inversion and (68) give

$$D = \frac{T}{\pi} \sum_{n \le X} a_n^2 g(y_n) + O(L^2 \log L). \tag{87}$$

For product frequencies,

$$\left| \int_{T+x_-}^{2T-x_+} e^{i\tau' \log(nm)}\, d\tau' \right| \le \frac{2}{\log(nm)} \le \frac{2}{\log 4}.$$

Returning to the expression before the x-integration and taking absolute values therefore gives

$$\begin{aligned} |O_2| &\le \frac{1}{2\pi^2} \sum_{n,m\le X} a_n a_m \int_{-T}^{T} \Phi(x)^2 \frac{2}{\log(nm)}\, dx \\ &\le \frac{1}{\pi^2 \log 4} \left(\sum_{n\le X} a_n \right)^2 \int_{\mathbb{R}} \Phi(x)^2\, dx. \end{aligned}$$

By the prime-power estimate

$$\sum_{n\le X} a_n = \sum_{n\le X} \frac{\Lambda(n)}{\sqrt{n}} \ll \sqrt{X}$$

and Plancherel

$$\int_{\mathbb{R}} \Phi(x)^2\, dx = 2\pi b L \ll L.$$

Consequently

$$\boxed{O_2 \ll XL.} \tag{88}$$

For $n \ne m$, put $\vartheta = y_n - y_m$ and

$$\alpha_n^+ = \int_0^T \Phi(x)^2 e^{ixy_n}\, dx, \qquad \alpha_n^- = \int_{-T}^0 \Phi(x)^2 e^{ixy_n}\, dx.$$

Then $|\alpha_n^\pm| \le \pi b L \ll L$. Direct evaluation of the inner τ'-integral gives

$$\begin{aligned} O_1 = \frac{1}{2\pi^2} \Re \sum_{n\ne m} \frac{a_n a_m}{i(y_n - y_m)} \Bigg[\left(\frac{n}{m}\right)^{2iT} (\alpha_m^+ + \alpha_n^-) \\ - \left(\frac{n}{m}\right)^{iT} (\alpha_n^+ + \alpha_m^-) \Bigg]. \end{aligned}$$

Thus O_1 is the sum of four Hilbert forms with coefficient pairs

$$(a_n n^{2iT}, a_m m^{-2iT} \alpha_m^+), \quad (a_n n^{2iT} \alpha_n^-, a_m m^{-2iT}),$$
$$(a_n n^{iT} \alpha_n^+, a_m m^{-iT}), \quad (a_n n^{iT}, a_m m^{-iT} \alpha_m^-).$$

We use the weighted Montgomery–Vaughan Hilbert inequality

$$\left| \sum_{r\ne s} \frac{x_r z_s}{\lambda_r - \lambda_s} \right| \ll \left(\sum_r \frac{|x_r|^2}{\delta_r} \right)^{1/2} \left(\sum_r \frac{|z_r|^2}{\delta_r} \right)^{1/2}, \qquad \delta_r = \min_{s\ne r} |\lambda_r - \lambda_s|,$$

with an absolute implied constant; no sharp constant is needed. For $\lambda_n = \log n$, $\delta_n^{-1} \ll n$, since adjacent integers already have logarithmic spacing $\gg 1/n$. Hence

$$\sum_{n\le X} \frac{a_n^2}{\delta_n} \ll \sum_{n\le X} \Lambda(n)^2 \ll XL,$$

and a sequence carrying $\alpha_n^{\pm}$ has the corresponding weighted square sum $\ll L^3 X$. Each of the four Hilbert forms is therefore $O(L^2 X)$, so

$$O_1 \ll L^2 X. \tag{89}$$

Combining (87)–(89),

$$\mathcal{M}[P_X, P_X] = \frac{T}{\pi} \sum_{n \le X} \frac{\Lambda(n)^2}{n} g(\log n) + O(L^2 X). \tag{90}$$

By Stieltjes partial summation from (71), and using $\|g'\|_\infty \le \|(\phi^2)'\|_1 = O(1)$,

$$\sum_{n \le X} \frac{\Lambda(n)^2}{n} g(\log n) = \int_0^L y g(y)\, dy + O(L^2). \tag{91}$$

C.5.3 Mixed and pole terms

Proposition C.4 (Secondary terms)**.** *At $L = \ell$ and $X \asymp T$,*

$$\mathcal{M}[\mu, P_X] \ll \ell\sqrt{X}, \qquad \mathcal{M}[\mu, \Pi_X] \ll \ell L\sqrt{X},$$

$$\mathcal{M}[P_X, \Pi_X] \ll LX, \qquad \mathcal{M}[\Pi_X, \Pi_X] \ll \frac{LX}{T}.$$

All four are $o(TL\ell^2)$.

Proof. Define

$$m(\tau') = \int_I \Phi(\tau - \tau')^2 \mu(\tau)\, d\tau.$$

Then $\|m\|_\infty \ll \ell L$. Differentiating under the integral and integrating by parts in τ gives

$$\int_I |m'(\tau')|\, d\tau' \ll \ell L,$$

using $\mu' = O(T^{-1})$ and (69). Therefore, for $y \ge \log 2$,

$$\left| \int_I m(\tau') e^{iy\tau'} d\tau' \right| \ll \frac{\ell L}{y}.$$

Substitution of the defining formula for P_X and $\sum_{n \le X} \Lambda(n)/(\sqrt{n}\log n) \ll \sqrt{X}/L$ yields $\mathcal{M}[\mu, P_X] \ll \ell\sqrt{X}$.

For the remaining terms use on I

$$|\Pi_X| \ll \frac{\sqrt{X}}{T}, \qquad 0 < \mu \ll \ell, \qquad |P_X| \ll \sqrt{X},$$

and

$$\sup_\tau \int_I \Phi(\tau - \tau')^2 d\tau' \le 2\pi b L \ll L.$$

Thus

$$|\mathcal{M}[\mu, \Pi_X]| \ll T \cdot \ell \cdot \frac{\sqrt{X}}{T} \cdot L = \ell L \sqrt{X},$$

$$|\mathcal{M}[P_X, \Pi_X]| \ll T \cdot \sqrt{X} \cdot \frac{\sqrt{X}}{T} \cdot L = LX,$$

and

$$|\mathcal{M}[\Pi_X, \Pi_X]| \ll T \cdot \frac{X}{T^2} \cdot L = \frac{LX}{T}.$$

□

C.6 Scale-free limit

Assume the fixed-width tapered family satisfies

$$\phi_L(Ls)^2 = v(s) + o(1)$$

in $L^1([-1/2, 1/2])$, with v bounded and supported on $[-1/2, 1/2]$. Then

$$a \to A := \int_{-1/2}^{1/2} v(s)\, ds, \qquad b \to B_v := \int_{-1/2}^{1/2} v(s)^2\, ds.$$

Writing

$$G_v(r) = \int v(s)v(s-r)\, ds,$$

one has $g(Lr) = LG_v(r) + o(L)$ in L^1, and hence

$$\frac{2}{L^3}\int_0^L yg(y)\, dy \to J_v := \iint_{[-1/2,1/2]^2} |s-t|v(s)v(t)\, ds\, dt. \tag{92}$$

Combining Proposition C.3, (86), (90), (91), Proposition C.4, and (92),

$$\operatorname{tr} \widetilde{G}^2 = \frac{TL}{2\pi}\Big(B_v\ell_1^2 + L^2 J_v\Big) + o(TL\ell^2).$$

Since, by (62),

$$N(T, 2T) = \frac{T\ell_1}{2\pi} + O(\ell), \qquad \frac{L}{\ell_1} \to 1,$$

and $\widehat{G} = \widetilde{G}/(aL)$,

$$\boxed{\|\widehat{G}\|_{\mathrm{F}}^2 = \left(\frac{B_v + J_v}{A^2} + o(1)\right) N(T, 2T).} \tag{93}$$

Thus

$$c_1(v) = \frac{A^2}{B_v + J_v}.$$

Together with (77), this proves the two required prime-side moments for every admissible profile.

C.7 Closing the Montgomery–Taylor optimization

On $H = L^2([-1/2, 1/2])$ define the self-adjoint integral operator

$$(\mathcal{T}v)(s) = \int_{-1/2}^{1/2} |s-t|v(t)\, dt.$$

Then

$$c_1(v) = \frac{\langle 1, v\rangle^2}{\langle v, (I+\mathcal{T})v\rangle}.$$

Lemma C.5 (Positivity and invertibility)**.** *$I + \mathcal{T}$ is strictly positive and invertible on H.*

Proof. Schur's test gives

$$\|\mathcal{T}\|_{2\to 2} \le \sup_{|s|\le 1/2} \int_{-1/2}^{1/2} |s-t|\,dt = \sup_{|s|\le 1/2} \left(s^2 + \frac{1}{4}\right) = \frac{1}{2}.$$

Hence

$$\langle v, (I+\mathcal{T})v\rangle \ge (1 - \|\mathcal{T}\|)\|v\|_2^2 \ge \frac{1}{2}\|v\|_2^2.$$

□

Cauchy–Schwarz in the positive inner product $\langle f, g\rangle_A = \langle f, (I+\mathcal{T})g\rangle$ now gives

$$c_1(v) \le \langle 1, (I+\mathcal{T})^{-1}1\rangle,$$

with equality precisely when

$$v \propto (I+\mathcal{T})^{-1}1.$$

Thus a maximizer satisfies

$$(I+\mathcal{T})v = C \tag{94}$$

for some constant C. In the interior,

$$(\mathcal{T}v)'' = 2v,$$

so (94) implies

$$v'' + 2v = 0.$$

By symmetry the maximizing solution is even, hence

$$v(s) = A_0 \cos(\sqrt{2}s).$$

Conversely, for $v_*(s) = \cos(\sqrt{2}s)$ the function $F = v_* + \mathcal{T}v_*$ satisfies $F'' = 0$ and is even, hence is constant; therefore v_* really satisfies (94), not merely its differentiated equation. Since $1/\sqrt{2} < \pi/2$, $v_* > 0$ on the whole interval, so the constraint $v \ge 0$ is inactive.

Let $\theta = 1/\sqrt{2}$. Then

$$A = \int_{-1/2}^{1/2} v_*(s)\,ds = \sqrt{2}\sin\theta.$$

Evaluating $C = (I+\mathcal{T})v_*$ at $s = 0$ gives

$$C = 1 + 2\int_0^{1/2} t\cos(\sqrt{2}t)\,dt = \cos\theta + \frac{\sin\theta}{\sqrt{2}}.$$

Since $\langle v_*, (I+\mathcal{T})v_*\rangle = CA$,

$$c_1^* = \frac{A}{C} = \frac{2\tan(1/\sqrt{2})}{\sqrt{2} + \tan(1/\sqrt{2})},$$

and therefore

$$\boxed{\frac{1}{c_1^*} = \frac{1}{2} + \frac{1}{\sqrt{2}}\cot\frac{1}{\sqrt{2}}.} \tag{95}$$

D Self-contained derivation of Input IV: Poisson–Gabor sampling

This appendix proves the critical-lattice Poisson identity, the tapered Fourier decay, finite-grid truncation, and the Montgomery–Taylor overlap limit. The window is real and even, so the real-axis transforms used below are real and even; modulus squares may be retained throughout.

D.1 Statement to be verified

Let $L \to \infty$, put

$$h = \frac{2\pi}{L}, \qquad \tau_k = kh,$$

and let $\phi_L \in C_c^2(\mathbb{R})$ be a real tapered window with exact support $[-L/2, L/2]$. Write

$$\widehat{\phi}_L(\xi) = \int_{\mathbb{R}} \phi_L(u) e^{-i\xi u}\, du$$

and

$$a_L = \frac{1}{L} \int_{\mathbb{R}} \phi_L(u)^2\, du.$$

For the Montgomery–Taylor choice, away from an $O(1)$ physical taper layer,

$$\phi_L(u)^2 = \cos\left(\frac{\sqrt{2}\, u}{L}\right), \qquad |u| \le \frac{L}{2} - O(1).$$

Define

$$K(x) = \int_{-1/2}^{1/2} \cos(\sqrt{2}\, t) \cos(2\pi x t)\, dt, \qquad k(x) = \frac{K(x)}{K(0)}.$$

The required form of Input IV is the following.

Theorem D.1 (Self-contained Input IV)**.** *Fix $R > 0$. Let γ, γ' be points whose normalized coordinates $x = L\gamma/(2\pi)$ and $x' = L\gamma'/(2\pi)$ are at least L^2 grid cells from both ends of the finite sampling interval, and suppose $|x - x'| \le R$. Then the normalized finite-grid Gram entry satisfies*

$$\frac{1}{a_L L^2} \sum_{\tau_j \text{ in the finite grid}} \widehat{\phi}_L(\gamma - \tau_j) \overline{\widehat{\phi}_L(\gamma' - \tau_j)} = k(x - x') + o(1),$$

uniformly for $|x - x'| \le R$.

D.2 The infinite-grid identity

Lemma D.2 (Critical-grid Poisson identity)**.** *Assume $\phi \in L^2(\mathbb{R})$ is compactly supported in an interval of length at most L and is sufficiently regular for Poisson summation. Then*

$$\sum_{k \in \mathbb{Z}} \widehat{\phi}(\tau - \tau_k) \overline{\widehat{\phi}(\tau' - \tau_k)} = L \int_{\mathbb{R}} |\phi(u)|^2 e^{-i(\tau - \tau')u}\, du.$$

Proof. Set

$$F_{\tau,\tau'}(\xi) = \widehat{\phi}(\tau - \xi)\, \overline{\widehat{\phi}(\tau' - \xi)}.$$

With the Fourier convention above, a direct Fourier inversion calculation gives

$$\widehat{F}_{\tau,\tau'}(s) = 2\pi e^{-i\tau' s} \int_{\mathbb{R}} \phi(u) \overline{\phi(u - s)} e^{-i(\tau - \tau')u}\, du.$$

Poisson summation on the lattice $h\mathbb{Z}$, $h = 2\pi/L$, gives

$$\sum_{k \in \mathbb{Z}} F_{\tau,\tau'}(kh) = \frac{1}{h} \sum_{m \in \mathbb{Z}} \widehat{F}_{\tau,\tau'}(mL).$$

Because ϕ is supported in an interval of length at most L, the translated supports of $\phi(u)$ and $\phi(u - mL)$ are disjoint for $m \neq 0$, apart from endpoints of measure zero. Thus only $m = 0$ survives. Since $1/h = L/(2\pi)$,

$$\sum_{k\in\mathbb{Z}} F_{\tau,\tau'}(kh) = L \int_{\mathbb{R}} |\phi(u)|^2 e^{-i(\tau-\tau')u} \, du.$$

□

Remark D.3. *The exact support condition* $\operatorname{supp} \phi_L = [-L/2, L/2]$ *is the one used by the optimized fixed-width tapered family. Hence the critical-density Poisson identity has no aliasing term: shifts by* mL*,* $m \neq 0$*, overlap only on sets of measure zero.*

D.3 Fourier decay

Lemma D.4 (Uniform decay)**.** *Suppose* $\phi_L \in C_c^2([-L/2, L/2])$ *is uniformly bounded and its taper has fixed physical width with uniformly bounded first and second derivatives. Then*

$$|\widehat{\phi}_L(r)| \ll \min\left\{L, \frac{1}{|r|}, \frac{1}{r^2}\right\},$$

with an implied constant independent of L*.*

Proof. The L^1 estimate gives $|\widehat{\phi}_L(r)| \ll L$. One integration by parts, using compact support and the bounded total variation of ϕ_L, gives $|\widehat{\phi}_L(r)| \ll |r|^{-1}$. A second integration by parts on each smooth piece gives $|\widehat{\phi}_L(r)| \ll |r|^{-2}$: the bulk derivatives of the Montgomery–Taylor profile contribute $O(L^{-1})$ and $O(L^{-2})$ per unit length, while the fixed-width taper contributes $O(1)$ total variation at each derivative level. All boundary terms vanish because the taper is compactly supported and C^2. □

D.4 Finite-grid truncation

Lemma D.5 (Pointwise tail)**.** *Fix* $R > 0$*. Suppose* γ, γ' *are at least* L^2 *grid cells from a finite-grid endpoint and* $|L(\gamma - \gamma')/(2\pi)| \leq R$*. Then the contribution of the omitted half-grid beyond that endpoint is* $O(L^{-2})$ *before Gram normalization.*

Proof. Let the omitted index be $n \geq L^2$ cells beyond the nearer endpoint. Since $h = 2\pi/L$ and the normalized separation of γ, γ' is bounded,

$$|\gamma - \tau_j| \asymp \frac{n}{L}, \qquad |\gamma' - \tau_j| \asymp \frac{n}{L}$$

uniformly in the stated region. Lemma D.4 therefore gives

$$|\widehat{\phi}_L(\gamma - \tau_j)\widehat{\phi}_L(\gamma' - \tau_j)| \ll \frac{L^4}{n^4}.$$

Consequently

$$\sum_{n\geq L^2} \frac{L^4}{n^4} \ll L^4 \int_{L^2-1}^{\infty} t^{-4} \, dt = O(L^{-2}).$$

The same estimate applies at the other endpoint. □

D.5 Montgomery–Taylor overlap limit

Lemma D.6 (Normalization and overlap)**.** *For the fixed-width tapered Montgomery–Taylor family,*

$$a_L = K(0) + o(1), \qquad K(0) = \int_{-1/2}^{1/2} \cos(\sqrt{2}t)\,dt = \sqrt{2}\sin(1/\sqrt{2}) > 0.$$

Moreover, uniformly for $|x| \le R$,

$$\frac{1}{a_L L}\int_{\mathbb{R}} \phi_L(u)^2 e^{-i(2\pi x/L)u}\,du = k(x) + o(1).$$

Proof. Put $u = Lt$. Outside a set of t-measure $O(L^{-1})$ arising from the fixed-width taper,

$$\phi_L(Lt)^2 = \cos(\sqrt{2}t)$$

on $[-1/2, 1/2]$. Uniform boundedness of the window makes the taper contribution $O(L^{-1})$ after normalization. Dominated convergence, uniformly for $|x| \le R$, gives

$$a_L = \int_{-1/2}^{1/2} \cos(\sqrt{2}t)\,dt + o(1) = K(0) + o(1)$$

and

$$\frac{1}{L}\int \phi_L(u)^2 e^{-i(2\pi x/L)u}\,du = \int_{-1/2}^{1/2} \cos(\sqrt{2}t)e^{-i2\pi xt}\,dt + o(1).$$

The odd imaginary part vanishes. Dividing by $a_L = K(0) + o(1)$ proves the claim. □

The overlap has the explicit form

$$K(x) = \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}},$$

with removable singularities interpreted by continuity.

Proof of Theorem D.1. Apply Lemma D.2 to the infinite grid. By Lemma D.5, replacing the infinite grid by the finite central grid changes the unnormalized sum by $O(L^{-2})$. Since $a_L \asymp 1$, division by $a_L L^2$ makes this error $O(L^{-4})$. The remaining infinite-grid expression, after normalization, is the overlap in Lemma D.6. This gives $k(x - x') + o(1)$ uniformly on every fixed compact set of normalized separations. □

D.6 Audit notes and exact scope

The proof above is independent of the zeta explicit formula, zero density, and prime-side pair correlation. It uses only:

1. Poisson summation on the critical lattice $2\pi\mathbb{Z}/L$;
2. compact support and regularity of the chosen window;
3. the fixed-width taper;
4. elementary integration by parts.

Thus Input IV is proved independently from Poisson summation and elementary Fourier estimates for the exact support/taper convention used in the main manuscript.

E Certificate metadata

Code and certificate availability. The complete Arb/FLINT verifier, the archived 256-bit certification run, machine-readable parameters, execution logs, environment information, SHA-256 hashes, and reproduction instructions are publicly available in the reproducibility artifact [21]. The frozen snapshot used for the present certificate is GitHub release `v1.0.0-paper`, attached to commit `c57f53e` and identified additionally by the verifier and log SHA-256 values below. The release contains the manuscript snapshot and the computational evidence used to certify Proposition 5.2. Subsequent changes to the repository's default branch are not part of the trust base of this certificate.

The frozen archival verifier in release `v1.0.0-paper` is

`certify_nonuniform_3900_v1_1.py`.

The file with the same path on the current default branch contains the v19 fail-closed cutoff assertion and is therefore intentionally not byte-identical to this frozen object. The SHA-256 of the frozen verifier is
`8eb7b4a17da8e114881264d3c2fc8daf262a0558d4248692b386c0fca2cc4301`.
The 256-bit log SHA-256 is
`4b0e62dcb5c4014504981f16e431144e3a01184b2aa2aa6b0bf650705d59db83`.

Field	Archived 256-bit run
verified	`true`
target	$F_{6,\text{nonuniform}} \geq 39/10000$
grid	4000
precision	256 bits
nodes	1,119,372
pruned	560,334
splits	559,038
maximum depth	39
initial boxes	1296
interval prunes	364,112
tangent prunes	190,953
pressure prunes	5,269
elapsed time	201.487482 s

The pressure vector recorded by the verifier is

$$(2714, 3733, 3553, 3553, 3733, 2714)/10^7.$$

The kernel-table SHA-256 is
`f75a3968f6daf7ea74867e430dc69fdca4276c4bb09f8cecc1b9846e8942c3f5`.
The second-derivative-table SHA-256 is
`f878d03d2eeca7fd34e3064a8dacb70d054fe3b3e9e3890c79a1f8297633cfb3`.
The archived environment reports CPython 3.14.7 and `python-flint==0.9.0`.

F Exact arithmetic

For the main refinement,

$$m = 450, \qquad A = \frac{39}{10000}(m-6) = \frac{4329}{2500}, \qquad Q = \frac{1}{500}(m-6) = \frac{111}{125}.$$

Hence
$$\frac{A}{m} = \frac{481}{125000}, \qquad \frac{Q}{m} = \frac{37}{18750}, \qquad 1 - \frac{A}{m} = \frac{124519}{125000}.$$
Therefore
$$\frac{H_{\mathrm{MT}} - Q/m}{1 - A/m} = \frac{450H_{\mathrm{MT}} - 111/125}{450 - 4329/2500} = \frac{1125000H_{\mathrm{MT}} - 2220}{1120671}.$$
The scalar adjacent-pair input is
$$g_0 = \frac{89}{100}, \qquad t = \frac{33}{2}, \qquad \frac{171389}{1000000} < k(g_0),$$
which implies
$$w(g_0) > \left(\frac{171389}{1000000}\right)^2 > \frac{2937}{100000} = t\beta g_0.$$
The contradiction margin is the positive rational
$$g_0 Q + \left(1 - \frac{1}{t}\right)\frac{450}{449} - A = \frac{6363}{30868750} > 0.$$
The accompanying script `verify_improved_bound.py` verifies these identities and rigorously encloses the transcendental quantities entering the final decimal bound.

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